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Commercial real estate markets after the end of “low for long”: risks and policy challenges

Caroline Roulet

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ABSTRACT/RÉSUMÉ

Commercial real estate markets after the end of "low for long": risks and policy challenges

The paper provides a comprehensive assessment of commercial real estate market conditions. It assesses the scale and sources of commercial real estate financing, highlighting the role of bank loans and the expansion of non-bank funding. The report describes debt redemptions, sustainability challenges, and the deteriorating credit profile of property developers. It discusses financial stability risks from commercial real estate exposures and identifies challenges in risk assessment and macroprudential policies, offering policy recommendations to strengthen global financial resilience.

JEL: G21, G23, G32, R33.

Keywords: Commercial Real Estate, Non-bank Financial Institutions, Financial Risk and Risk Management, Macroprudential Policy.

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Les marchés de l'immobilier commercial après une période de taux d'intérêt historiquement bas: risques et défis politiques

Ce document fournit une évaluation complète des conditions du marché de l'immobilier commercial. Il évalue l'ampleur et les sources de financement dans le secteur, notamment le rôle des prêts bancaires et l'expansion du financement non-bancaire. Le rapport étudie les échéances de remboursement de la dette dans le secteur, les défis en matière de durabilité et la détérioration du profil de crédit des promoteurs. Il examine les risques pour la stabilité financière liés aux expositions au secteur de l'immobilier commercial et identifie les défis en matière d'évaluation des risques et de politiques macroprudentielles, en proposant des recommandations politiques pour renforcer la résilience financière mondiale.

JEL: G21, G23, G32, R33.

Mots clés: Immobilier Commercial, Institutions Financières Non-Bancaires, Risques Financiers et Gestion des Risques, Politiques Macroprudentielles.

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Commercial real estate markets after the end of "low for long": risks and policy challenges

By Caroline Roulet[1]

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Overview

Real estate assets are the largest store of wealth in the OECD countries, valued at USD 111 trillion in 2022, nearly twice the level of GDP. Commercial buildings, plus other corporate structures, at an estimated USD 38 trillion, account for more than one-third of total property assets, with the remainder residential real estate. After significant increases, commercial property prices have recently declined, particularly in the United States and the euro area, driven by both higher global interest rates and the adjustment to new demand patterns induced by growing teleworking and e-commerce. Since 2022, the financing costs of property developers and risk premia for low-rated commercial mortgage-backed securities have risen. The performance of commercial real estate investment trusts and funds has also weakened, with significant outflows from large US-based funds.

In 2023, the commercial real estate exposures of financial intermediaries are estimated to have totalled USD 21 trillion. Although loans collateralized by commercial property assets and loans to property developers account for a sizeable proportion of total bank lending, particularly in advanced economies, commercial real estate financing is largely provided by non-bank financial intermediaries. Prior to the Global Financial Crisis (GFC), commercial real estate financing was primarily from commercial mortgage-backed securities and direct loans from, and securities issuance to, banks and institutional investors. Over the past fifteen years, commercial equity and mortgage real estate investment trusts that either invest directly in property or hold property loans have significantly expanded, particularly in advanced economies and in the retail, office and industrial property sectors. Debt accumulation by commercial real estate investment trusts and property developers is reflected in a substantial increase in outstanding bonds and syndicated loans, and a growing reliance on private capital.

Cross-country information on the different aspects of commercial property financing is sparse. This paper provides an overview of commercial real estate financing and exposures by aggregating available data on different market-based instruments. It highlights a number of key challenges, risks, and policy implications from the ongoing downturn in the sector. Key financing-related issues include:

- Commercial property developers face mounting refinancing and credit quality challenges. Over the next five years, USD 1.3 trillion in bonds and syndicated loans will mature, representing 40% of the amount outstanding in 2023. Most redemptions will be from commercial property developers, mainly in China and non-US advanced economies.
- Debt sustainability challenges may increase for leveraged issuers, with the amount of low-credit-quality debt of the riskiest property developers now surpassing GFC levels. Mortgage credit quality is also deteriorating, as reflected by rising delinquency rates on commercial mortgage-backed securities and number of missed mortgage payments from borrowers with commercial real estate loans.
- Deteriorating credit quality in the commercial real estate sector could pose significant risks to financial stability, due to the large exposures of various financial intermediaries to the sector. Banks' non-performing loans and potential losses for non-bank financial intermediaries could rise due to increasing financing costs and pressure on the earnings of commercial property developers. Significant losses at commercial real estate investment trusts could spread to real estate investment funds, raising investor redemptions and potentially downward price spirals from fire sales that spill over into the wider financial sector. Insurance companies' solvency ratios and pension funds' liquidity may also be adversely affected by rising losses, both directly from their commercial real estate loans and indirectly through their interconnectedness with real estate funds.

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Policy challenges remain in assessing and mitigating risks from commercial real estate exposures. Significant data gaps hinder accurate analysis of financial stability risks from those exposures across financial intermediaries. Moreover, existing macroprudential tools in commercial real estate finance are primarily bank-based, with limited measures available to monitor non-bank financial intermediaries. The use of these tools can also vary across jurisdictions. There are a number of policy options that would help in monitoring and mitigating financial stability risks from the commercial real estate sector. These include:

- The monitoring of financial stability risks would be facilitated by cross-country agreement on a precise definition of commercial property assets and the development of common qualitative and quantitative indicators.
- Increased evaluation of the credit quality and loss-provisioning of commercial real estate portfolios held by both banks and non-bank financial intermediaries is needed, along with enhanced banking sector stress-testing by supervisors (with account taken of indirect exposures to commercial real estate through bank funding of exposed non-bank financial intermediaries). These measures would help to identify the major vulnerabilities of interconnected providers of commercial real estate credit that could pose systemic risk concerns, and would provide a stronger basis for policy recommendations.
- Improved assessment and mitigation of liquidity mismatches and leverage risks for institutional investors and real estate investment funds, including by constraining redemptions in the latter, would help to strengthen financial resilience.
- Further work on borrower-based measures (especially for non-bank financial intermediaries) and enhanced international cooperation would help to limit excessive risk-taking and regulatory fragmentation by encouraging a broad approach to risk monitoring and supervision, and facilitating the implementation of activity-based regulatory measures.

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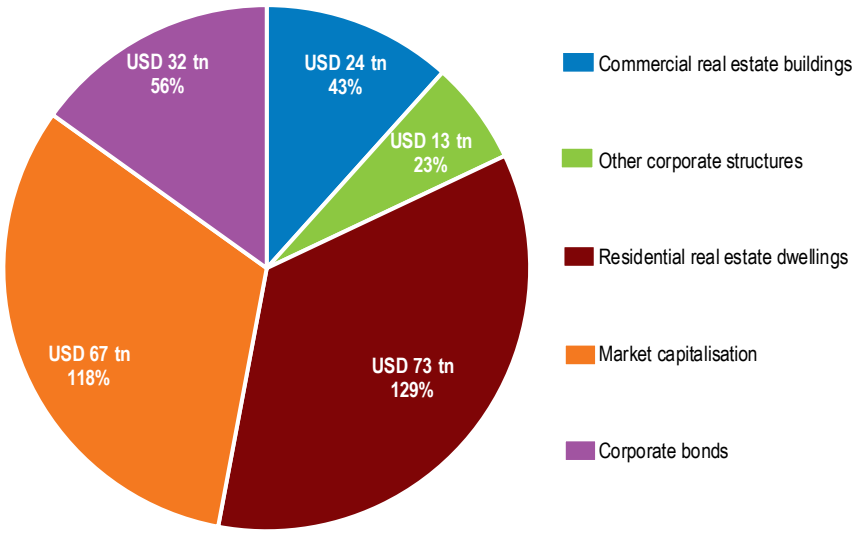
Commercial real estate market conditions have deteriorated

This section provides a comprehensive assessment of commercial real estate market conditions, including property valuations, financing costs, risk premia associated with commercial real estate investments, and the performance of real estate investment trusts (REITs) and funds.[2] The value of tangible real estate assets (excluding agricultural land) in the OECD countries reached USD 111 trillion in 2022, accounting for more than half of total selected marketable corporate financial assets and nearly twice the level of GDP (196%, Figure 1). Commercial real estate buildings and other corporate structures represented an estimated 22% and 12% of total real estate value respectively.

Figure 1. Value of real estate and selected marketable corporate financial assets in 2022

Aggregate of 34 OECD countries, USD trillion and % of GDP



Note: Based on data for 34 OECD countries (all members except Chile, Colombia, Costa Rica, and Mexico). Capital stock data are expressed in USD trillion as of 2022 and a share of GDP. Real estate capital stock data are sourced from the OECD annual balance sheet for non-financial assets database. The value of the commercial property stock is measured using the net value of "buildings other than dwellings" in the private financial and non-financial business sectors, which is a measure of productive buildings held in the private corporate sector that contribute to economic output. The net value is calculated as the gross value less the value of consumption of fixed capital (depreciation). The value of the commercial properties other than buildings is measured using the net value of "other structures" in the private financial and non-financial business sectors, which is a measure of the costs of infrastructure building and maintenance related to private corporate sector activities. The value of the residential property stock is measured using the net value of "dwellings", which is a measure of housing properties owned by households, general government sector, non-profits serving households and private corporations. For comparison, data for equity market capitalisation and corporate bonds (including financial and non-financial corporates) are presented for the same group of economies.
Source: Bank for International Settlements, OECD National Accounts, World Bank WDI, OECD calculations.

Commercial property prices in the United States and the euro area have declined significantly since their peak in mid-2022 (by 11% and 13% respectively; Figure 2 Panel A), though with significant heterogeneity across different property segments. The commercial real estate sector is adjusting to both higher interest rates and reshaped demand brought about by global trends such as the rise in teleworking and e-commerce. In the United States, prices in the office and retail property sectors have declined, with increasing use of remote work and a shift away from bricks-and-mortar shops to online retailing. In contrast,

[2] The definition of real estate investment trusts and funds is provided in Box 1.

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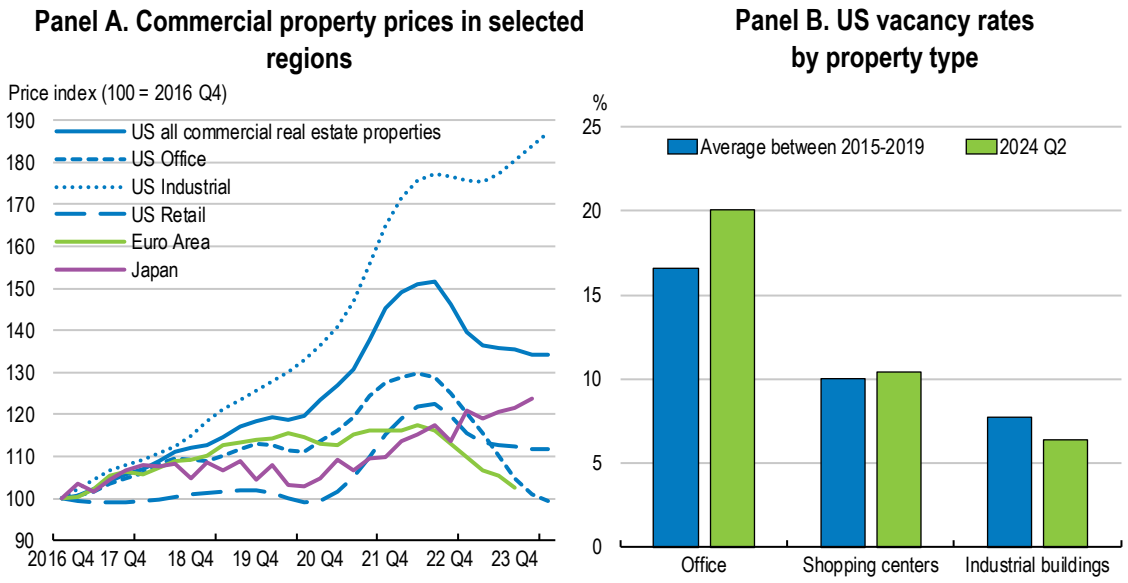
industrial property has benefitted from growing demand for e-commerce-related logistics platforms (JP Morgan, 2024[1]). Vacancy rates have tended to mirror developments in prices (Figure 2 Panel B), though occupancy rates in US retail shopping centres have been supported by demand from new types of tenants (Forbes, 2024[2]).

Figure 2. Commercial property prices in selected advanced markets and US vacancy rates



Note: In panel A, MSCI commercial property price indices measure the actual price movements for the main US commercial properties (including office, industrial and retail) based on exclusive transaction data using repeat-sales regression methodology. Nominal commercial property price indices used for the euro area and Japan are commercial properties (i.e. commercial land, offices, retail premises and industrial) rented out at market prices, built for future sale, or used for the production of market goods and services. In panel B, US vacancy rates for selected commercial properties are derived from Moody's Analytics Real Estate Information Services indicators. Data are presented as percentages of commercial properties' built space currently unoccupied or available for rent.
Source: LSEG, Bank for International Settlements, OECD calculations.

The decline in commercial property prices has coincided with an increase in financing costs for property developers since mid-2022 (Figure 3, Panel A), bringing them closer to expected returns (measured by the capitalisation rate[3]) at the time of property purchase. Option-adjusted spreads[4] of the lowest rated investment-grade US commercial mortgage-backed securities have also widened, with investors demanding a higher risk premium amid heightened credit risk (Figure 3, Panel B). Commercial real estate prices could still fall further, particularly for segments suffering from depressed demand, oversupply and elevated refinancing costs, such as the market for central business district offices (IMF, 2024[3]). A risk is that deeper price corrections could make the return on commercial property investments fall below the cost of financing commercial property purchases with debt, potentially exposing investors in these segments to losses.

[3] The capitalisation rate is the rate of return on a real estate property investment based on the income that the property is expected to generate.

[4] The option-adjusted spread measures the difference in yield between a bond with an embedded option (such as an mortgage-backed security or a callable bond) and a sovereign bond. Embedded options are provisions included with some fixed-income securities that allow the investor or the issuer to do specific actions, such as calling back the issue.

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Significant price corrections have already occurred for listed commercial REITs (Figure 4, Panel A), particularly for US commercial mortgage REITs (mREITs), US and European office REITs and European and Asian retail REITs, with some values declining by more than 50% between 2020 and 2024 (Figure 4, Panel B). Box 1 provides an overview of these different types of non-bank intermediaries that have become increasingly important in commercial property markets.

Box 1. Real estate non-bank financial intermediaries and market-based credit products

Real estate non-bank financial intermediaries

Real estate investment trusts

Within the real estate investment trust (REIT) industry, there are two broad types of trusts: equity REITs and mortgage REITs (mREITs):

- Equity REITs invest in physical properties and manage income-producing real estate that typically generates earnings through rents. Most trusts specialise in a single property type, although some manage portfolios that include multiple types of properties. Commercial equity REITs specialise in commercial real estate, and residential equity REITs in residential real estate.
- mREITs invest in mortgages and mortgage-backed securities (MBS), making them real estate debt owners. Commercial mREITs typically focus on commercial mortgages and commercial MBS, while residential mREITs invest in residential mortgages and residential MBS. mREITs engage in a wide range of activities related to originating, funding, servicing, restructuring and securitising loans secured by commercial real estate. Some mREITs specialise in one or two of these activities, while others perform several of them. Some mREITs also hold physical residential or commercial properties as investments. mREITs earn income from the interest and capital gains on their investments.

REITs use leverage. Equity REITs initially use the cash raised from investors to purchase tangible properties. Similarly, mREITs use the cash they raised from investors to purchase mortgage-backed securities. Then, REITs use tangible properties or mortgage-backed securities as collateral to borrow money to purchase more real estate assets, a process that they repeat multiple times. However, the number of rounds is limited as the (repo) lender requires a collateral margin on each loan, a gap that serves as a buffer for the lender's protection, or may impose covenants with regard to leverage. REITs also rely on unsecured revolving credit facilities and short-term credit to manage temporary liquidity needs.

Real estate investment funds

A real estate investment fund is a type of mutual fund that pools money from multiple investors to primarily invest in securities offered by real estate companies, including mortgage-backed securities, bonds and REITs shares.

While REITs pay out regular dividends or coupons, real estate investment funds reward investors through gains in asset values. Since most trusts are listed and traded on major stock exchanges, they tend to be more liquid than mutual investment fund shares, which can only be redeemed at the end of the trading day when the net asset value is settled.

The option-adjusted spread helps investors compare cash flows from a fixed-income security to reference rates while also valuing embedded options against general market volatility.

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Real estate market-based credit products

Real estate mortgage-backed securities

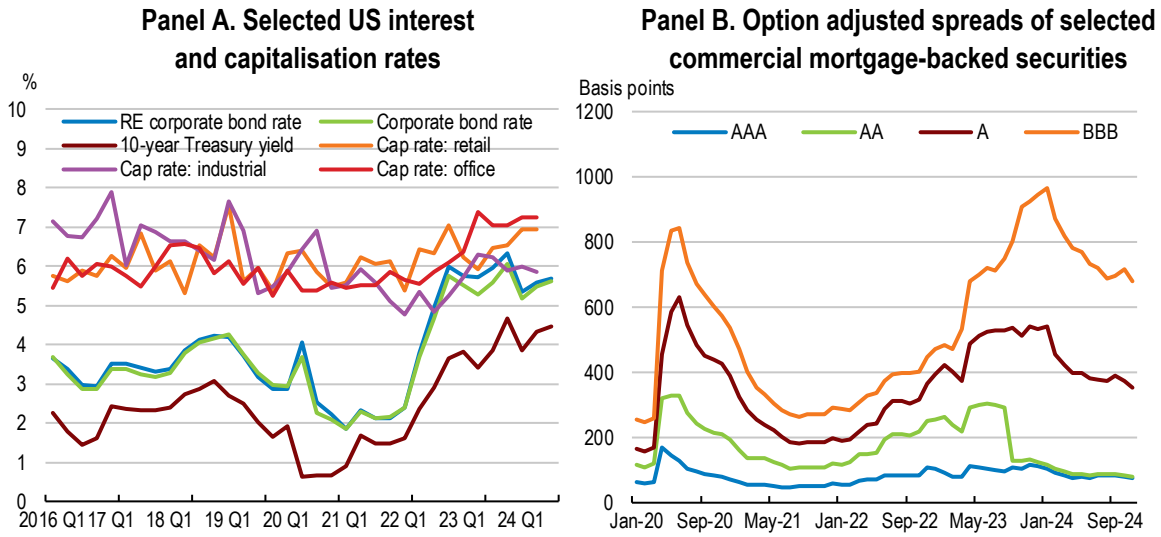
There are two types of real estate mortgage-backed securities (MBS): residential and commercial. Such mortgage-backed securities are bonds backed by a pool of commercial mortgages or residential real estate mortgages. In the United States, commercial mortgage-backed securities can also be collateralised by multi-family mortgages.

Bonds and syndicated loans

While most bonds issued by property developers are standard ones, sustainable bonds form a specific category that includes green, social, sustainability and sustainability-linked bonds. Green bonds enable capital-raising and investment for new or existing real estate projects with environmental benefits, including those aimed at fostering a net-zero emissions economy. Social bonds raise funds for new and existing socially sound and sustainable real estate projects (such as social housing). Sustainability bonds finance or re-finance a combination of both green and social real estate projects.

A syndicated loan is a commercial loan provided by a group of lenders to a property developer, initially structured and arranged by one or more commercial or investment banks, and then syndicated to other banks or institutional investors. Syndicated loans are rated by credit quality as either investment-grade or leveraged (speculative-grade) loans.

Figure 3. Selected financing costs, commercial property capitalisation rates and mortgage-backed securities' spreads



Note: In panel A, financing costs correspond to corporate bond yields of property developers and non-financial corporates, which are derived from ICE BofA global corporate bond indices, and ten-year US Treasury yield. Capitalisation rates (also known as cap rates) are derived from US Bloomberg indicators by selected commercial property type. The capitalisation rate measures the rate of return on a real estate investment property based on the income that the property is expected to generate. In panel B, Bloomberg commercial mortgage-backed securities (CMBS) investment grade indices cover US agency and non-agency CMBS markets. These indices are only available for US CMBS and not for markets in other jurisdictions. Option-adjusted spread (OAS) is the number of basis points that is added to the semi-annually compounded forward one-month rate curve in order to match the present value of cash flows to the bond's price. For CMBS indices, OAS are the calculated spreads between a computed OAS index of CMBS in a given rating category and a spot Treasury curve.
Source: Bloomberg, LSEG, OECD calculations.

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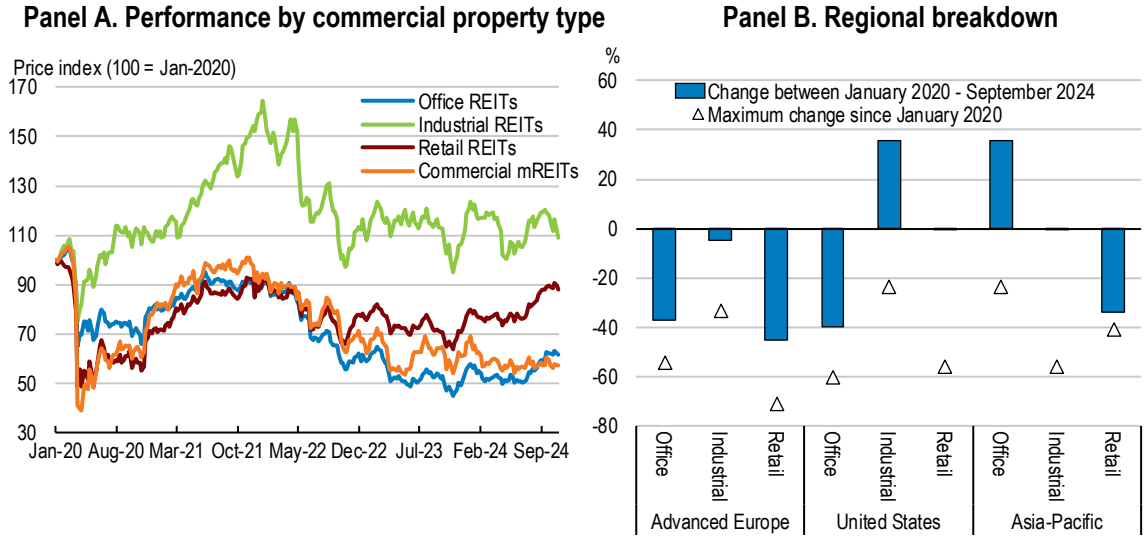
Figure 4. Performance of real estate investment trusts in selected advanced markets



Note: In panel A, REITs' performance is derived from selected FTSE EPRA Nareit indices by commercial property type in advanced markets (i.e. in North America, Europe and Asia-Pacific). The commercial mREITs index contains all US REITs that invest in commercial mortgages and CMBS. The European Public Real Estate Association (EPRA), in collaboration with the Financial Times Stock Exchange (FTSE) and the National Association of Real Estate Investment Trusts (Nareit), is the provider of the leading global listed real estate index. Panel B provides regional comparisons over the period from January 2020 to July 2024.
Source: LSEG, OECD calculations.

Tighter financial conditions and the deteriorating outlook in certain commercial real estate segments have also contributed to the weaker post-pandemic performance of real estate exchange traded funds (ETFs)[5] compared to diversified developed markets stock market ETFs (Figure 5, Panel A). Flows to real estate ETFs have also been volatile since 2020 (Figure 5, Panel B).

A particular concern is that the value of some portfolios of specialised investment funds may be overstated. Real estate investment trusts and funds revalue their commercial real estate investments on a regular basis but less frequently than for liquid assets, reflecting the lack of mark-to-market prices and the high costs of revaluing commercial properties. Therefore, trusts and funds may not have yet reported the full extent of the decline in the value of their commercial real estate investments.

[5] A real estate ETF is an investment fund that is traded on a stock exchange and tracks the performance of a specific real estate index or a basket of real estate securities.

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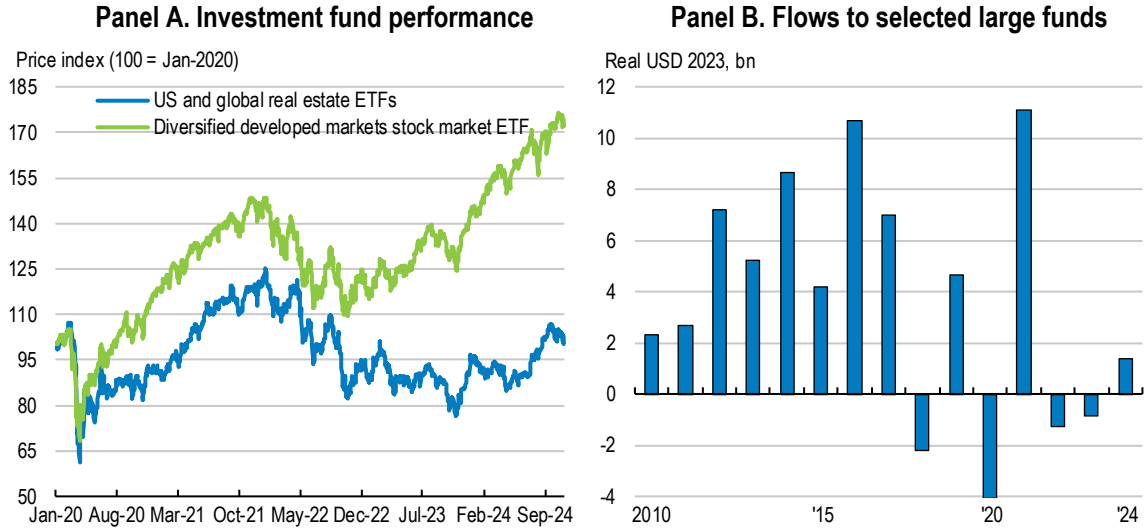
Figure 5. Performance and flows in real estate investment funds



Note: In panel A, the evolution of market prices of top six US and global real estate ETFs (Vanguard Real Estate ETF, Schwab US REIT ETF, Real Estate Select Sector SPDR Fund, iShares Global REIT ETF, Vanguard Global ex-U.S. Real Estate ETF, SPDR Dow Jones Global Real Estate ETF) is shown versus largest diversified developed markets stock market ETF (Vanguard FTSE Developed Markets ETF). In panel B, investment flows are shown for the top US and global real estate ETFs (including 6 funds) by assets under management in January 2024. Fund flow data are expressed in billion US dollar adjusted by 2023 US consumer price index. Data in 2024 cover the period from January, 1st to September, 9th.
Source: etf.com, LSEG, OECD calculations.

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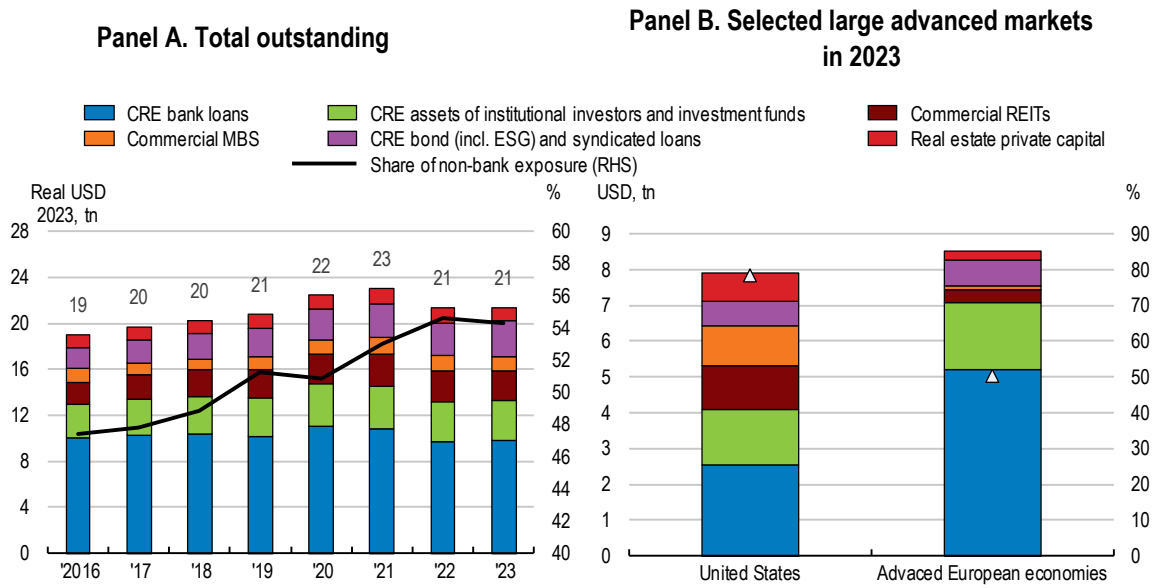
Exposures of financial intermediaries to commercial real estate

This section provides an overview of the different sources of commercial real estate funding and thus exposures to the sector, providing the background for the discussion of the ensuing risks in the following section. As commercial real estate investment statistics derived from national accounts suffer from various gaps, including limited data availability and a lack of comparability across countries, national accounts databases are supplemented by new datasets assembled for this paper. These datasets provide global coverage for various funding sources, including banks, non-bank financial intermediaries, and dedicated market-based products. Annex B provides methodological details on the different datasets that have been assembled and the calculations.

The aggregate commercial real estate exposures of financial intermediaries are estimated to have remained broadly steady over the past decade, reaching an estimated USD 21 trillion in 2023 (Figure 6, Panel A). Within this, the share of exposures from non-bank sources has risen over time and now exceeds 50%. The non-bank institutional sector is extremely diverse, especially in the United States (Figure 6, Panel B).

Figure 6. Commercial real estate exposures



Note: This figure provides estimates of commercial real estate exposures for both bank and non-bank financial intermediaries. Data are expressed in trillion US dollars adjusted by 2023 US consumer price index. Bank loans (blue bars) include loans collateralized by commercial property assets and loans to property developers. Real estate exposures of institutional investors and investment funds (green bars) include real estate property holdings (excluding own use), commercial real estate mortgages, commercial mortgage-backed securities (CMBS), and shares in real estate investment funds. These data are sourced from OECD and IMF financial accounts and national sources covering all available advanced and emerging economies. Real estate market-based finance includes all relevant securities and marketable assets available in LSEG across advanced and emerging economies (see Box 1 and Annex B for definitions and methodological details). Commercial REITs (brown bars) represent assets from a global sample of publicly traded commercial equity REITs and US commercial mortgage REITs (mREITs). Commercial MBS (orange bars) denotes outstanding amounts of commercial and diversified or specialised real estate mortgage-backed securities issued worldwide. Additionally, the global outstanding amount of corporate bonds issued by commercial and diversified or specialised property developers (including sustainable bonds), and bonds issued by commercial and diversified or specialised equity REITs and syndicated loans to property developers are included (purple bars). These data exclude residential property-only developers. Global assets under management of real estate private capital investment funds are shown in the red bars. Panel B shows the breakdown of real estate financing in the United States and advanced European economies only because data are available for these countries in all the underlying datasets.
Source: IMF FSI, OECD National Accounts, National central banks and office statistics, LSEG, Pitchbook, OECD calculations.

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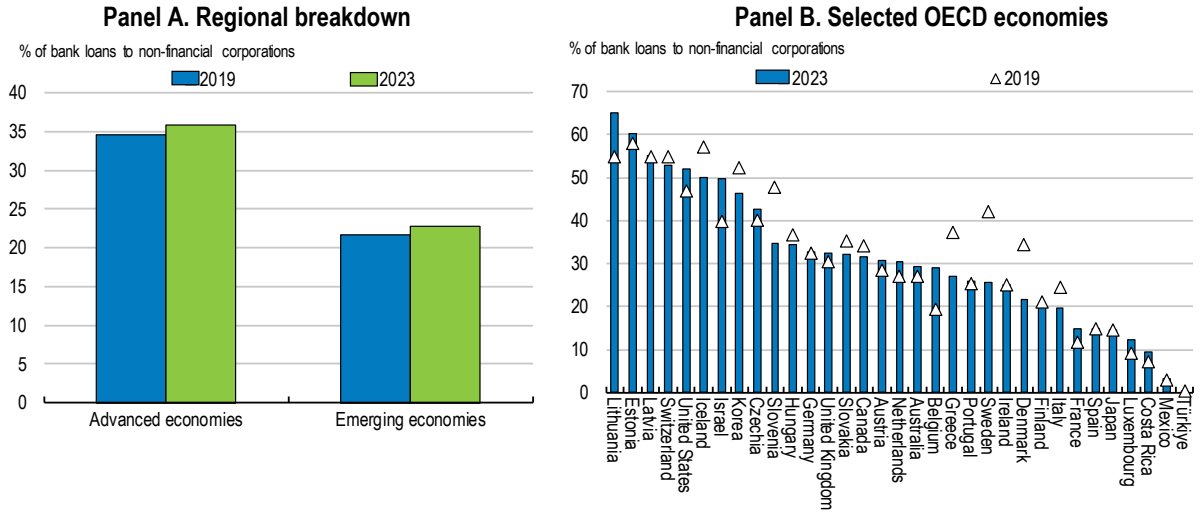
Banks still have significant commercial real estate loans

Commercial real estate loans account for a sizeable proportion of total bank loans to non-financial corporations, particularly in advanced economies (Figure 7, Panel A). However, there are notable differences across countries (Figure 7, Panel B). In the United States, smaller and regional banks have sharply increased the share of commercial real estate loans in total loans since the pandemic, while the largest banks have not (Figure 8, Panel A). With more than half of commercial real estate loans by both large and small US banks backed by commercial (i.e. non-farm non-residential) properties, they are particularly exposed to strained conditions in the office and retail segments (Figure 8, Panel B). In the euro area, banks' commercial real estate exposures are less than 100% of their equity, but several small banks have exposures that reach multiples of their equity. This makes them more vulnerable to downturns in the sector (EBA, EIOPA and ESMA, 2024[4]). Against this backdrop, banks in many countries could become increasingly reluctant to lend to firms that rely on commercial real estate assets as collateral while declines in commercial property prices persist.[6] Bank lending standards have also become more restrictive in many countries, making it more challenging for leveraged borrowers in the commercial real estate sector to secure fresh financing or roll over existing loans.

Figure 7. Commercial real estate loans as a share of bank loans to non-financial corporations in selected advanced and emerging economies



Note: In panel A, the average share of commercial real estate loans to total bank loans to non-financial corporations is shown using a sample of 31 advanced and 50 emerging economies where data on commercial real estate bank loans are available. Data for Russia are excluded. In European Union economies, data are sourced from the European Banking Authority. Commercial real estate loans are defined as the gross carrying amount of loans at amortised cost to non-financial corporations collateralised by commercial immovable property. In other economies, data are sourced from the IMF Financial Soundness Indicators database. Commercial real estate loans comprise both loans collateralised by commercial property assets and loans to construction companies and companies active in property development (including those companies involved in the development of multi-household dwellings and social housing). Under this definition, commercial properties include buildings, structures, and associated land used by enterprises for retail, wholesale, manufacturing, or other purposes (IMF, 2019[5]). More granular data showing loans collateralised by commercial property assets versus loans to property developers are not available. Given the different definition of commercial real estate loans in EBA and IMF data, comparisons between EU and non-EU countries should be regarded with prudence.
Source: EBA, ECB, IMF FSI, OECD calculations.

[6] Banks are exposed to commercial real estate markets through two main channels. First, credit risk on loans to firms exposed to the commercial real estate sector could increase following a market shock. Second, as commercial property assets are extensively accepted as collateral by banks, potential losses may increase following rising borrower defaults amid sharp declines in the value of commercial property collateral. The collateral channel may create a link between commercial property prices and wider credit conditions (Kiyotaki and Moore, 1997[48]; Bernanke, Gertler and Gilchrist, 1999[49]).

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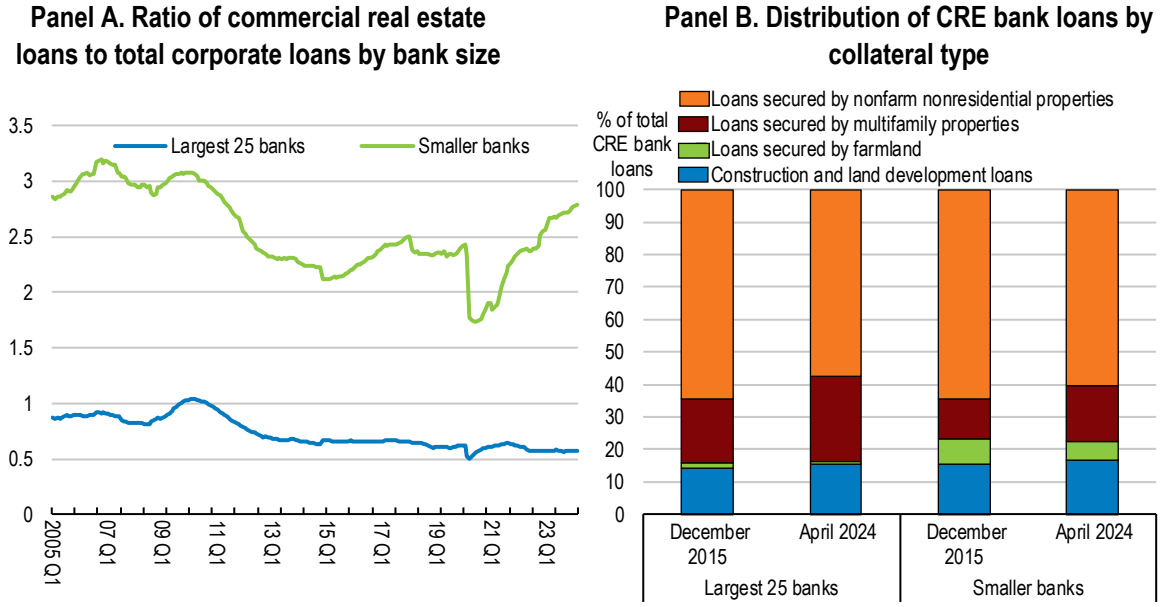
Figure 8. US banks' commercial real estate exposure by size and collateral type



Note: Commercial real estate loans comprise both loans collateralised by commercial property assets and loans to construction companies and companies active in property development (including those companies involved in the development of multi-household dwellings and social housing). Under this definition, commercial properties include buildings, structures, and associated land used by enterprises for retail, wholesale, manufacturing, or other purposes. Large domestically chartered commercial banks are defined as the top 25 domestically chartered commercial banks, ranked by domestic assets as of the previous commercial bank Call Report (to which the Federal Reserve's H.8 release data have been benchmarked). Small domestically chartered commercial banks are defined as all domestically chartered commercial banks not included in the top 25. The data for large and small domestically chartered banks are adjusted to remove the estimated effects of mergers and panel shifts between these two bank groups.
Souce: Federal Reserve, LSEG, OECD calculations.

Non-bank financing and dedicated market products have expanded significantly

Substantial commercial real estate financing from investment funds and institutional investors

Amongst major non-bank investors, investment funds have the highest shares of commercial real estate exposure in total assets, followed by insurance companies and pension funds, though there is considerable heterogeneity across jurisdictions (Figure 9). The amount of assets under management of real estate investment funds (one type of investment fund) has risen significantly over time (Figure 10).[7] This has offered long-term investment opportunities to other investors, such as life insurance companies and pension funds in Europe, leading to real estate investment funds becoming more interconnected with other financial institutions. However, the stable funding structures and long-term investment strategies of insurers and pension funds help to ensure that real estate funds have stable funding structures, reducing the risk of large-scale redemptions from open-ended funds.

[7] The substantial declines in 2022 and 2023 suggest that real estate investment funds maintain large investment portfolios sensitive to valuation effects.

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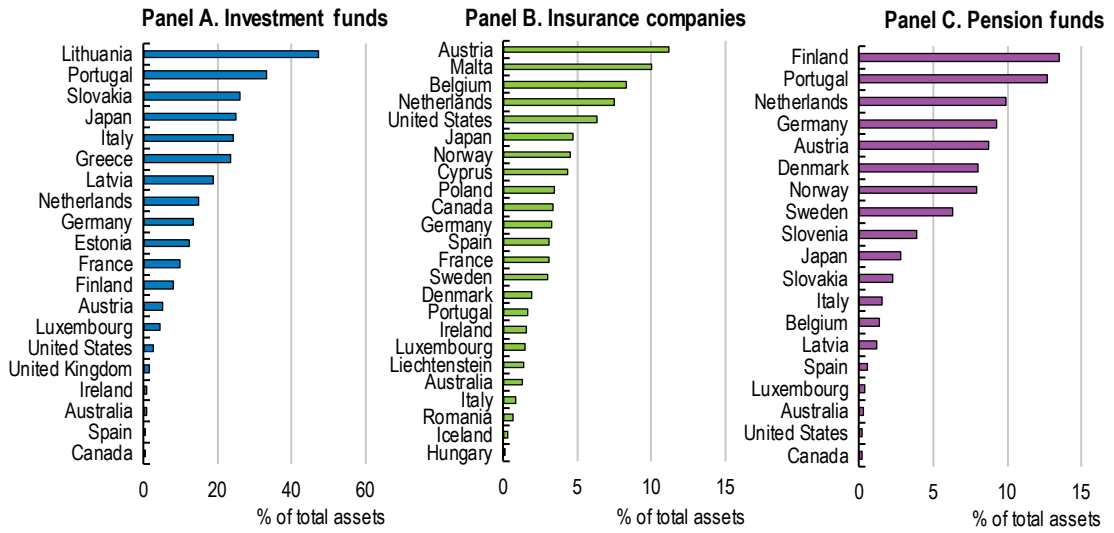
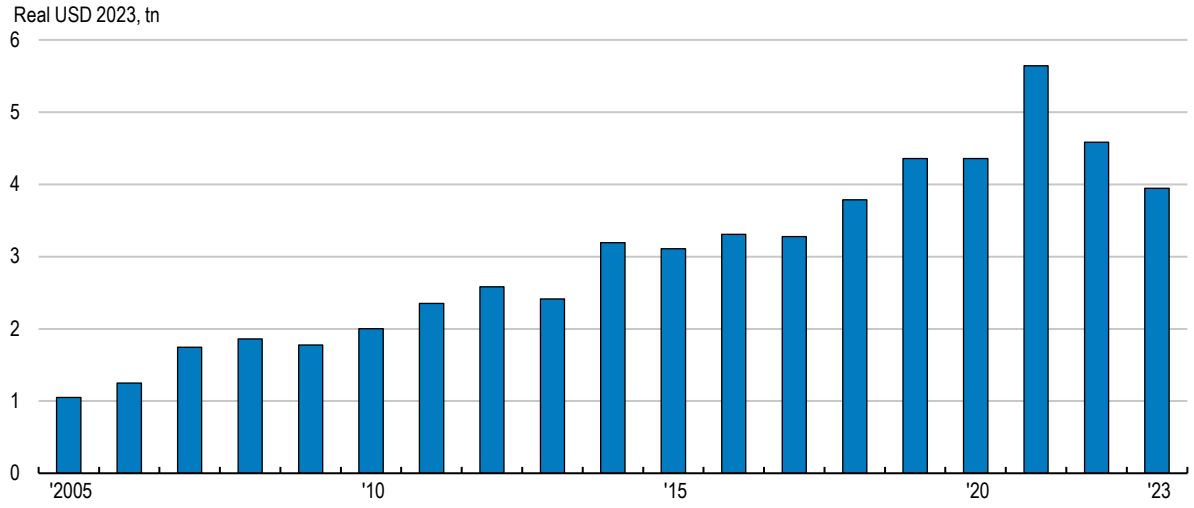
Figure 9. Commercial real estate exposure of investment funds and institutional investors



Note: Real estate exposures are proxied by real estate property holdings (not for own use), commercial real estate mortgages, commercial mortgage-backed securities and real estate investment funds shares. The investment fund data encompass the assets of real estate investment trusts and funds. The commercial real estate exposure of insurance companies and pension funds includes shares of real estate investment trusts and funds. Consequently, some assets of trusts and funds are reflected in the data for insurance companies and pension funds. Data are not available separately for insurance companies and pension funds in the United Kingdom. Data are as of 2023 and are expressed as a share of total assets.
Source: Australian Bureau of Statistics, Bank of Japan, EIOPA Insurance and Pension statistics, European Central Bank, Federal Reserve, UK Office for National Statistics, Statistics Canada, OECD calculations.

Figure 10. Global assets under management of real estate investment funds



Note: Data are adjusted by 2023 US consumer price index.
Source: The ANREV/INREV/NCREIF Fund Manager Survey 2024, OECD calculations.

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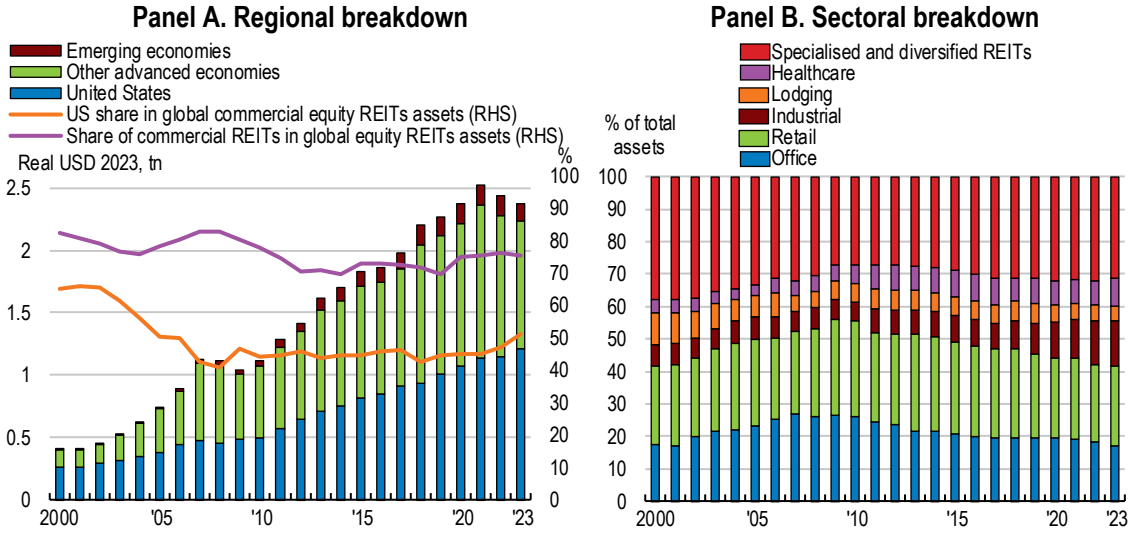
Commercial equity real estate investment trusts have grown significantly

Publicly traded commercial equity real estate investment trusts (REITs, see Box 1 for a definition) have expanded significantly over the past twenty years (Figure 11, Panel A).[8] REITs in the retail and office sectors, along with specialised and diversified REITs, account for the majority of REITs assets (Figure 11, Panel B). The exposures of the commercial equity REIT sector are overwhelmingly in advanced economies. Leveraged REITs in retail and office sectors, along with some specialised and diversified REITs, may be particularly sensitive to any sharp depreciation of retail and office property prices.

Figure 11. Total assets of publicly-traded commercial equity REITs by region and sector



Note: In panel A, the share of commercial equity REITs in the global total is the ratio of commercial equity REITs' assets in global equity REITs' assets (i.e. including commercial and residential equity REITs). Panel B is based on the subsample of commercial and diversified or specialised equity REITs. See Box 1 and Annex B for definitions and methodological details.
Source: LSEG, OECD calculations.

Commercial mortgage-backed securities markets have experienced a mild recovery since the GFC, and commercial mortgage REITs have expanded significantly

The global issuance of commercial mortgage-backed securities (MBS) has declined to a quarter of the pre-GFC level in real terms, with issuance almost equally split between the United States and other advanced economies. (Figure 12, Panel A). As a result, the global amount outstanding peaked in 2007 and has declined gradually since then (Figure 12, Panel B). More recently, a category of REIT that invests in commercial mortgages (mREITs)[9] have expanded significantly in the United States (Figure 12, Panel C). Exposures to commercial mortgage-backed securities are closely tied to commercial property market conditions in advanced economies. Consequently, the current downturn in the US retail and office segments and the euro area commercial property market could lead to credit quality deterioration and downgrades for commercial mortgage-backed securities.

[8] These are included in the assets of investment funds in Figure 9.

[9] See Box 1 for detailed definitions of equity and mREITs.

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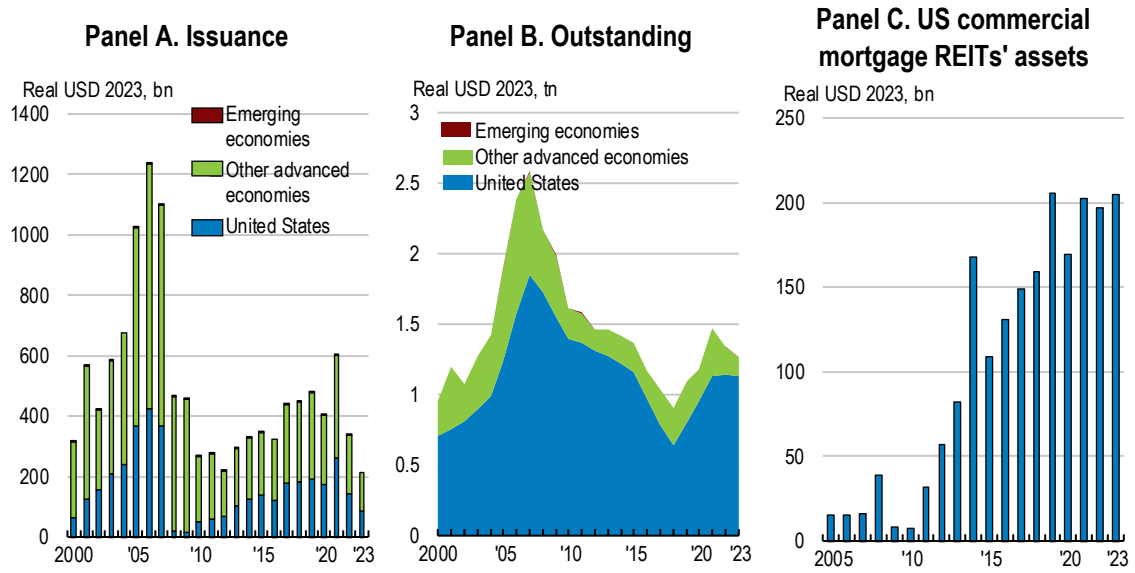
Figure 12. Commercial mortgage-backed securities and commercial mortgage REITs



Note: Panels A and B show the issuance and outstanding amounts of commercial mortgage backed securities globally. See Box 1 and Annex B for definitions and methodological details.
Source: LSEG, OECD calculations.

Commercial property developers and equity real estate investment trusts have increasingly resorted to bond issuance

Global bond issuance by commercial equity REITs and property developers surged up until 2021 (Figure 13, Panels A and B), leading to a growing outstanding amount and indebtedness in these sectors (Figure 14). Similar trends occurred in other sectors after the GFC, as non-financial companies around the world took advantage of low borrowing costs amid accommodative monetary policies (OECD, 2024[6]). Tighter banking regulations have also contributed to the growth of non-bank financing.

Commercial equity REITs bond issuance is dominated by US issuers. Conversely, bond issuance by commercial property developers' has increasingly been dominated by Chinese non-residential property developers, and issuers in non-US advanced economies. Commercial property developers have also issued a higher proportion of bonds at floating rates in emerging economies than in advanced economies. A sizeable share of bonds in emerging economies are also issued in foreign currency (Figure 13, Panel C). Thus commercial property developers in emerging economies are particularly sensitive to fluctuations in interest and exchange rates.

Corporate bond issuance by commercial equity REITs and property developers declined sharply in 2022 and 2023 (Figure 13, Panels A and B), in part reflecting the higher cost of debt. However, it may also reflect that many commercial equity REITs and property developers raised sufficient financing during the low-interest rate period. For such issuers the impact of higher interest rates will be felt gradually as existing bonds mature and new funding needs arise.

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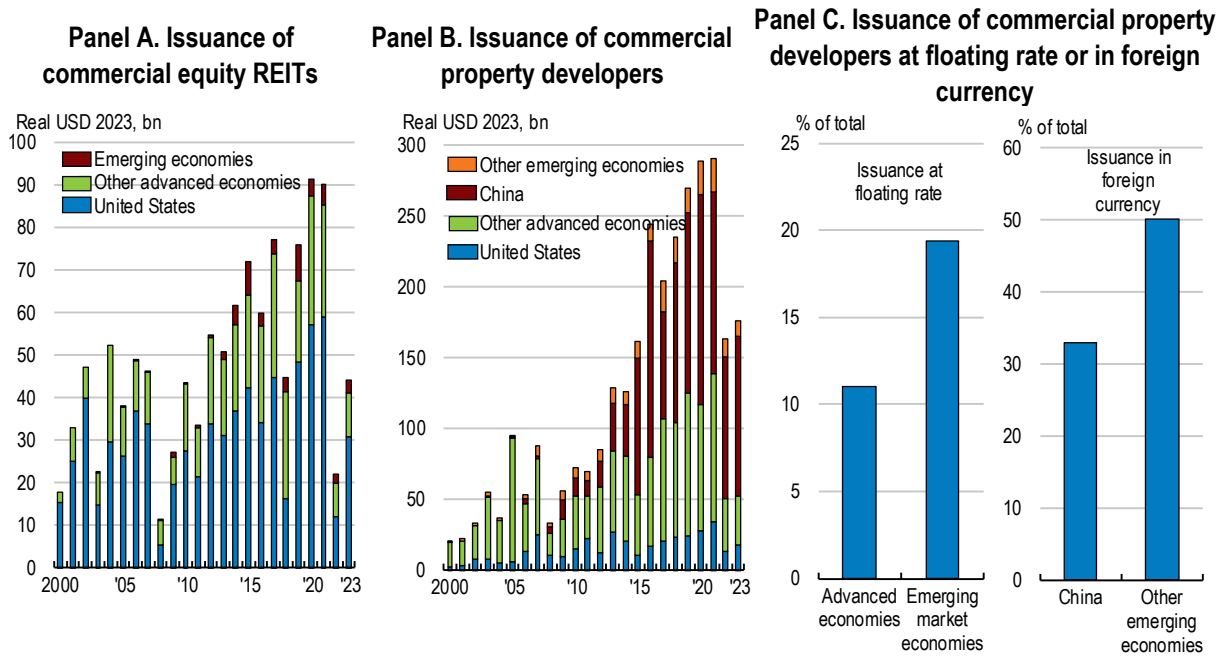
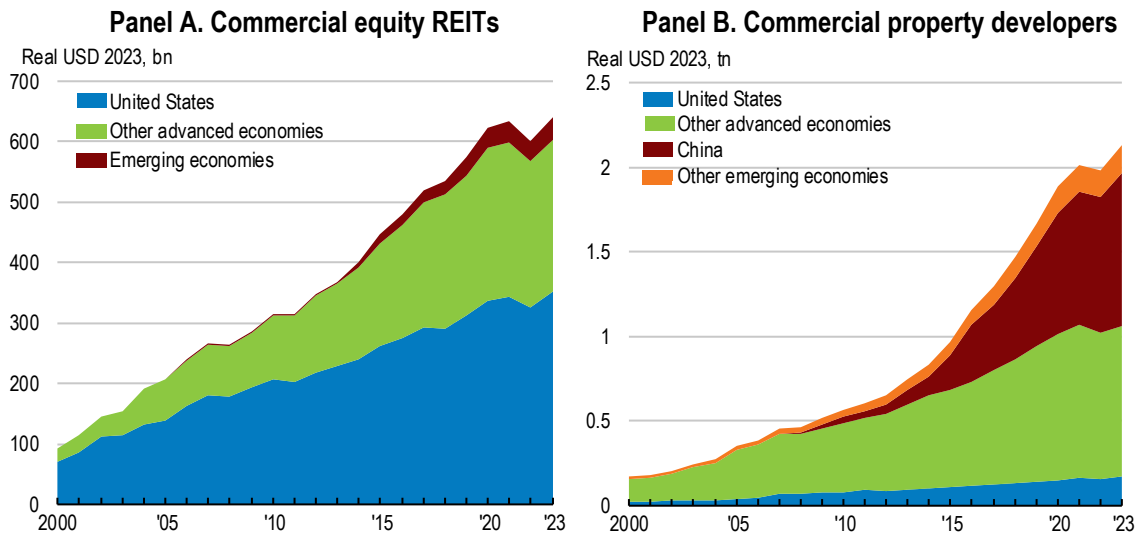
Figure 13. Global bond issuance of commercial equity REITs and property developers



Note: The bar charts in panel C depict the average share from 2014 to 2023 of bonds issued by commercial property developers at floating rates in advanced and emerging economies, and the share of bonds issued in foreign currency in China and other emerging economies. Domestic and US dollar bonds, along with syndicated loans, have grown rapidly for non-financial corporates in emerging economies over the past fifteen years (BIS, 2024[7]). In advanced economies, bonds are typically issued in domestic currencies, which are major international currencies. See Box 1 and Annex B for definitions and methodological details.
Source: LSEG, OECD calculations.

Figure 14. Outstanding bonds outstanding of commercial equity REITs and property developers



Note: See Box 1 and Annex B for definitions and methodological details.
Source: LSEG, OECD calculations.

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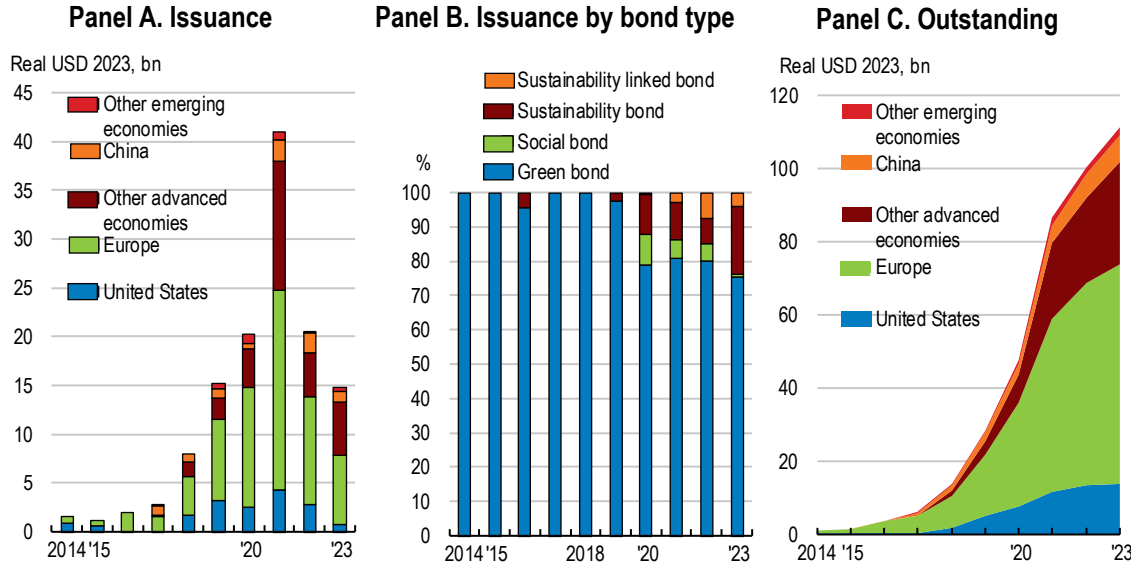
Issuance of green and sustainable bonds by commercial property developers has increased significantly over the past decade with many investors increasingly seeking to align their portfolios with internationally recognised sustainability goals (OECD, 2022[8]). Most issuance has occurred in Europe and other non-US advanced economies and largely comprises green bonds to finance green buildings (Figure 15, Panels A and B). Policy developments related to the European Green Deal have been the driving force behind the increased issuance of green bonds in Europe.[10] Recent regulatory and standard-setting initiatives by the OECD, G20 and Financial Stability Board members have also sought to encourage issuance by strengthening sustainability-related disclosures, the regulation of investment funds and the guidance for ESG[11] rating and data providers (OECD, 2024[9]). However, in practice commercial real estate sustainable bond issuance remains volatile, as reflected by the sharp declines recorded in 2022 and 2023, and commercial real estate sustainable bond markets remain relatively small compared to conventional corporate and syndicated bond markets for commercial property developers (Figure 15, Panel C and Figure 14, Panel B.

Figure 15. Global issuance and outstanding amount of commercial real estate sustainable bonds



Note: See Box 1 and Annex B for definitions and methodological details.

Syndicated loan markets for property developers have been resilient

Despite substantial declines in 2022-23, issuance levels in syndicated loan markets surpassed pre-pandemic values across all regions (Figure 16, Panels A and B), with the global outstanding amount of syndicated term loans reaching a new peak in 2023 (Figure 16, Panel C). Nevertheless, the amount outstanding is still only about one fifth of the outstanding corporate bonds of commercial property developers (Figure 14, Panel B). Challenging economic and financial conditions have recently resulted in deteriorating credit quality for many companies, and ratings downgrades for some property developers.

[10] The 'Fit for 55' package was initially proposed in July 2021 to respond to the requirements in the EU Climate Law, with a final legislative package adopted in 2023. This is designed to make the EU's climate, energy, transport and taxation policies fit for reducing net greenhouse gas emissions by at least 55% by 2030 and therefore make green bond investments attractive, compared to 1990 levels (European Commission, 2023[61]).

[11] ESG stands for environmental, social and governance.

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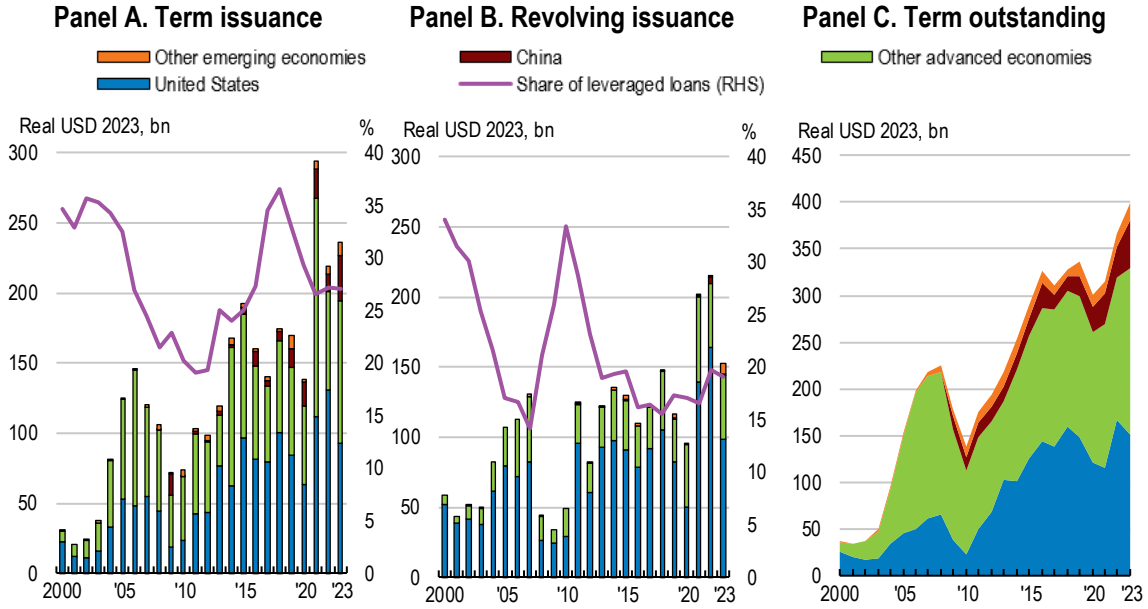
Together with the large fixed costs of issuing bonds, this has made access to corporate bond financing harder than access to syndicated loans for many borrowers.

Figure 16. Global issuance and outstanding amounts of syndicated loans to property developers



Note: See Box 1 and Annex B for definitions and methodological details.
Source: LSEG, OECD calculations.

Real estate private capital is a volatile financing source, particularly during times of uncertainty

The global assets under management of real estate private capital funds have doubled since 2010, although declines have been recorded since 2021 (Figure 17). This volatility, especially during periods of uncertainty, highlights changing investors' risk sentiment. It is also a potential source of financial instability given the opaqueness of private capital markets, which are not regulated like commercial banks or other traditional financial institutions (IMF, 2024[10]).

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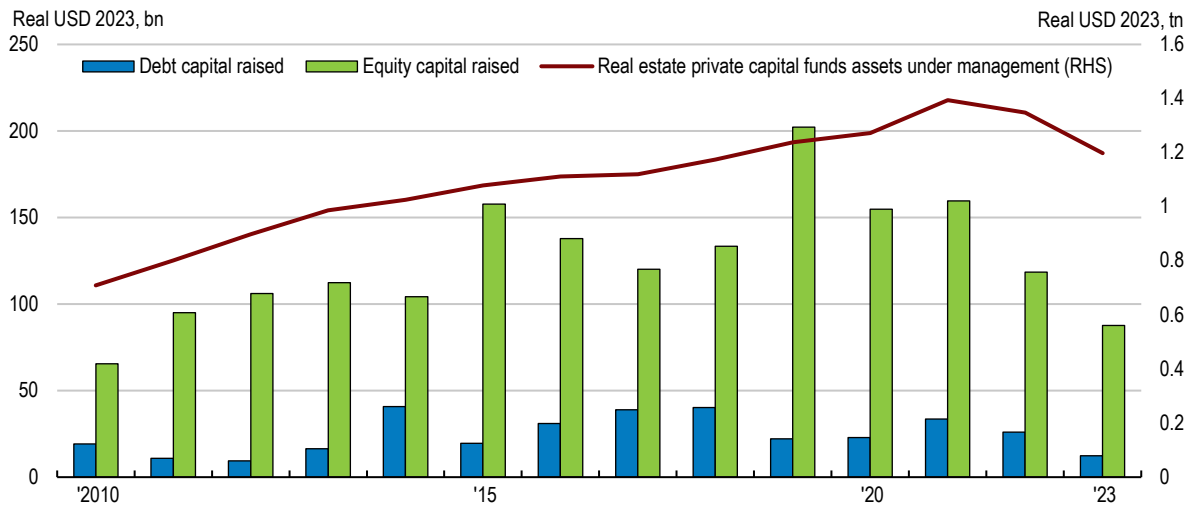
Figure 17. Real estate private capital raised and assets under management



Note: This figure shows capital (i.e. equity and debt) raised and assets under management of private capital real estate investment funds. Data are expressed in billion and trillion US dollars respectively adjusted by the US consumer price index.
Source: Pitchbook, OECD calculations.

Debt sustainability of commercial real estate borrowers and financial stability risks

This section analyses risks to financial stability from the multiple exposures to commercial real estate documented above. It first describes debt sustainability challenges that arise from the deteriorating credit profile of property developers and real estate investment trusts, the rise in commercial real estate mortgage delinquencies, and major forthcoming debt redemptions. It then discusses spillover risks across bank and non-bank financial intermediaries.

Commercial real estate borrowers have a deteriorating credit profile and face significant debt redemptions

Low credit-quality debt of property developers, based on the degree of leverage and interest coverage ratios of companies,[12] has increased substantially since 2020 and now exceeds GFC levels (Figure 18, Panel A). Developers in emerging economies[13] are particularly affected, but credit quality has also

[12] Credit quality assessment is based on leverage and repayment capacities, as rating information was unavailable for many bonds in the database.

[13] House prices rose significantly relative to household incomes in China pre-pandemic, in part because consumers preferred to invest their considerable savings in real estate given limited attractive alternative options. Property developers borrowed rapidly amid expectations of continuous home and land price increases, with land sales providing crucial revenue for local governments. To address overbuilding and rising construction sector debt, Chinese authorities implemented new regulations on debt limits for developers (known as the Three Red Lines). The Chinese property crisis began in 2021 with Evergrande's default on payments to international investors, followed by other large property developers becoming non-viable in subsequent years. While debt restructuring and liquidations of large developers occurred in the recent years, many bankruptcies have been avoided in part due to rules that allow lenders to delay recognizing their bad loans. This has helped mute spillovers to real estate prices and bank balance sheets (IMF, 2024[62]). Large state-owned banks appear to be solid, but problems could lurk among smaller banks dependent on the real estate sector and local governments (Bank of Finland, 2024[63]).

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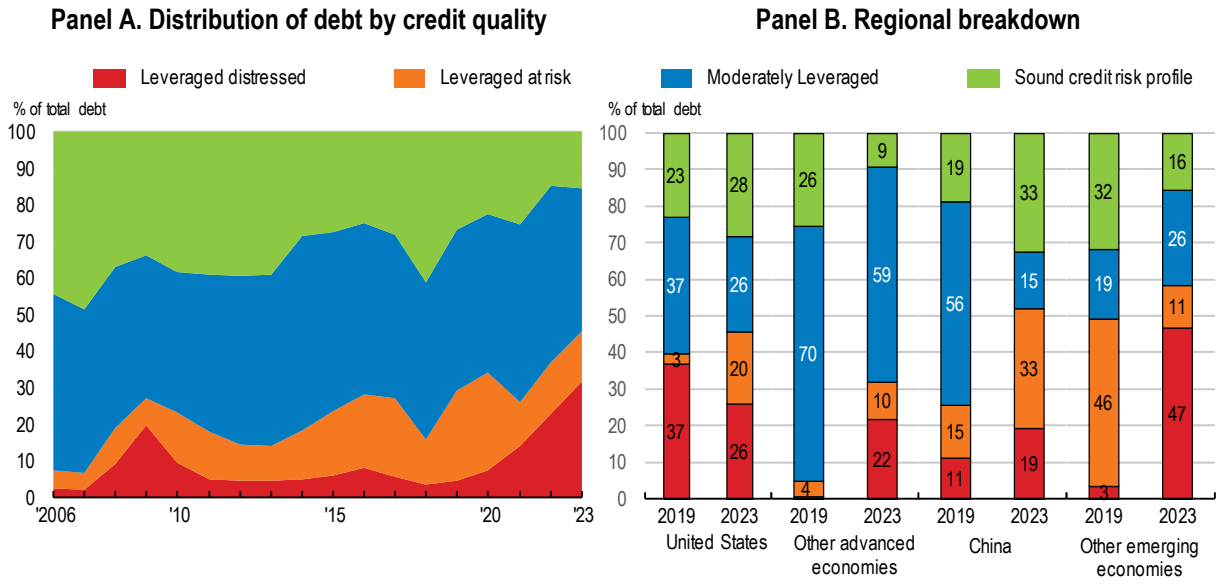
deteriorated in many advanced economies (Figure 18, Panel B). Low quality debt has also risen for commercial equity REITs, but to a much smaller extent. Within the commercial property sector, low credit-quality debt is primarily within the lodging, retail and office markets.[14] However, other than in the office sector, the share of low-quality debt has diminished from the peaks in 2020-2021.

Figure 18. The composition of property developers' debt



Note: Debt data include the total debt reported on the liability side of the balance sheet. See Annex B for definitions and methodological details.
Source: LSEG, OECD calculations.

A further sign of deteriorating credit quality is the rise in delinquency rates on commercial mortgage-backed securities, particularly in the US office sector (Figure 19, Panel A). Some borrowers with commercial real estate loans have also started to miss mortgage payments, as reflected by rising so called "early delinquent" mortgage ratios in some European economies, particularly Germany, Luxembourg and Belgium (Figure 19, Panel B). A potential concern is that credit rating agencies mis-rate commercial real estate investments (Financial Times, 2024[11]), as seen in the subprime loan crisis and the large mortgage-backed securities downgrades during the GFC. Credit agencies may take time to make downgrades to avoid the possibility of sending misleading signals about commercial real estate borrower credit quality that could potentially prompt fire sales of commercial mortgage-backed securities and tangible properties and a decline in investor confidence.

[14] Figures for commercial equity REITs are not shown but are available upon request.

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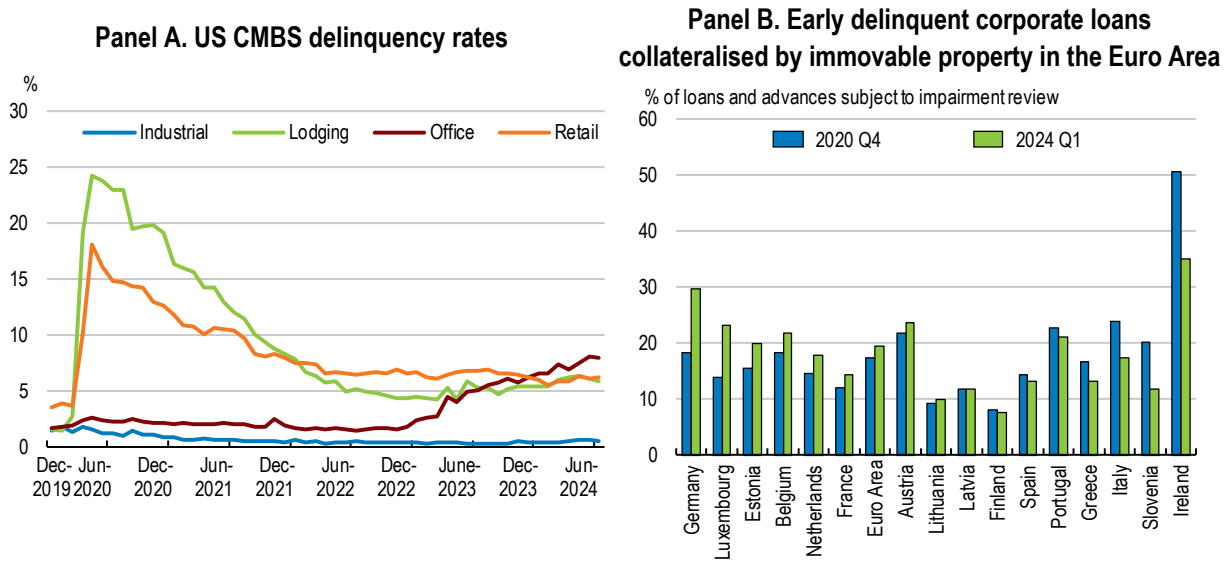
Figure 19. Delinquency rates have risen in some market sectors



Note: In Panel B, early delinquent corporate loans refers to stage 2 loans to non-financial corporations. Stage 2 assets under IFRS 9 accounting standards are financial instruments that have deteriorated significantly in credit quality since initial recognition but offer no objective evidence of a credit loss event. Countries are ordered by gap between 2024 and 2020 ratios.
Source: Trepp, European Central Bank.

In addition to maturing bonds and loans in 2024, a significant additional amount of commercial real estate bonds and syndicated loans is due to mature in the following four years. Globally, this amounts to USD 1.3 trillion, constituting 40% of the outstanding amount in 2023 (Figure 20, Panel A). The majority of these redemptions are from commercial property developers, mainly concentrated in China and non-US advanced economies (Figure 20, Panel B). Maturing syndicated loans and sustainable bonds are also mainly in advanced economies other than the United States. In contrast, redemptions of commercial equity REITs are primarily in the United States. Near-term redemptions are particularly substantial for syndicated loans of property developers, with an estimated 31% having become due in 2024 (Figure 20, Panel C).

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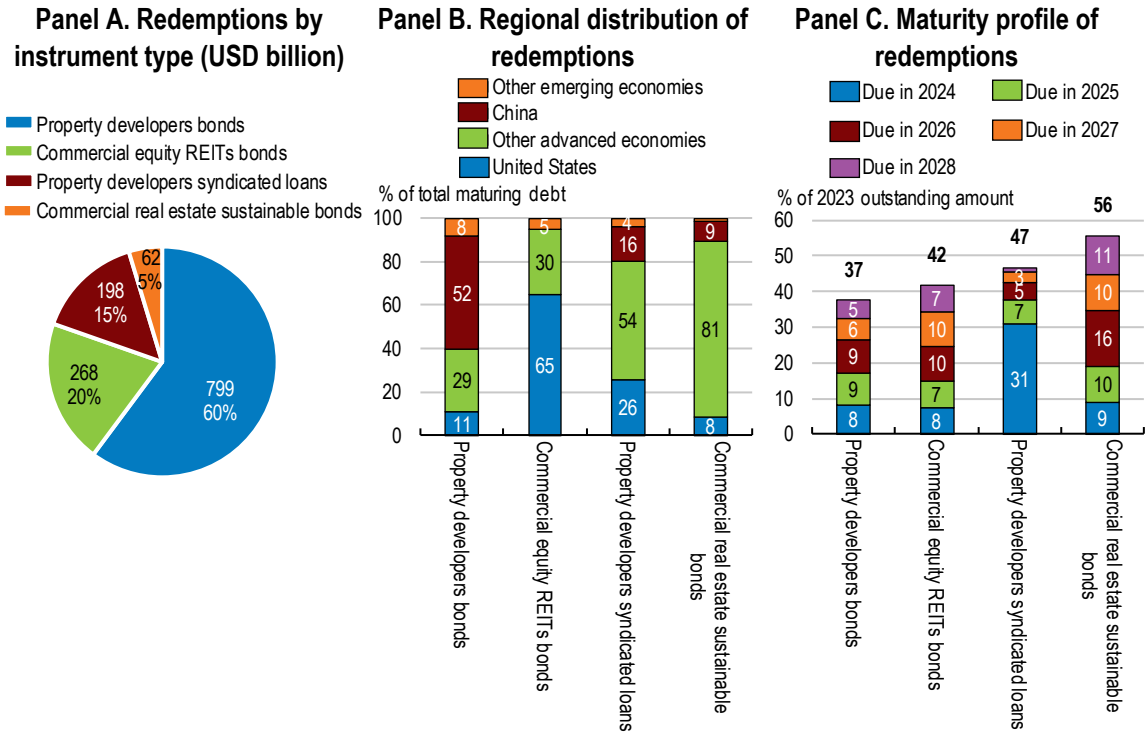
Figure 20. Redemptions of commercial real estate bonds and syndicated loans in 2024-28



Note: See Box 1 and Annex B for definitions and methodological details.
Source: LSEG, OECD calculations.

Debt sustainability challenges in leveraged commercial real estate finance would be exacerbated if tight financial conditions were to prompt institutional investors to reduce the risk of their portfolios. During the prolonged period of low interest rates following the GFC, institutional investors favoured high-yielding bonds and direct loans (including to property developers) to enhance their ability to fulfil future commitments. Tighter monetary policy has made bonds, particularly investment-grade rated corporate and government conventional bonds, more attractive to institutional investors. The performance of commercial property investments has also been eroded by declining property prices. A swift change in institutional investors' investment strategies would heighten funding challenges for leveraged commercial real estate issuers, and potentially raise delinquencies.

Risks to financial stability from the commercial real estate exposures of financial intermediaries

Spillover risks from deteriorating debt servicing capacities

One key risk is that bank non-performing loans and potential losses for non-bank financial intermediaries could rise due to increased financing costs and the weak profitability outlook for commercial property developers (European Central Bank, 2023[12]). Both factors threaten the debt servicing capacity of commercial property developers, compounded by falling sale prices, contracting order books, and shifts in lease revenues and office occupancy rates (Deghi, Natalucci and Qureshi, 2022[13]). Volatile energy prices and the ongoing transition to green real estate could also raise costs and capital expenditures to meet more demanding energy efficiency requirements for both developers and landlords (OECD, 2023[14]). Rising leverage and deteriorating interest coverage ratios may pose challenges to debt servicing capacity,

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with possible rating downgrades and defaults if some "at risk" firms fall into the "distressed" category. Downside risks in commercial real estate exposures of banks and non-bank financial intermediaries could also be exacerbated if large losses for commercial property coincided with stress in other sectors due to a weak corporate profitability outlook (OECD, 2024[15]).

Possible deleveraging and losses from fragilities at real estate investment trusts and funds and institutional investors

Declines in property prices and lower rents could also have an adverse impact on commercial equity REITs and mREITs. The former could face significant losses and the latter might be forced to reduce the net present value of their mortgages, leading in both cases to a contraction of their assets and deleveraging. A concern is that the share of lower credit-quality debt has risen (Figure 18) and the capacity to cover interest payments with cashflow[15] has deteriorated in 2022-2023 for many commercial equity REITs, particularly in lodging, retail and office property sectors. Potential refinancing challenges could force these institutions to deleverage,[16] amplifying price declines for tangible real estate assets and also commercial mortgage-backed securities given that mREITs are key buyers of such securities.

Strained financial conditions at commercial REITs could also spillover to real estate investment funds through significant losses on their holdings of REITs equity and subsequent redemptions from investment fund investors. In this event, fire sales of REIT shares might exceed the ability of dealers and other investors to absorb them. This would amplify downward price spirals in increasingly illiquid markets, transmit stress to other parts of the financial system, and possibly disrupt the availability of finance to the real economy.

Institutional investors are also exposed to possible rating downgrades of commercial real estate bonds and mortgage-backed securities, and subsequent losses from their direct commercial real estate loans and their interconnections with real estate investment funds. Potentially, this could have a negative impact on the solvency ratios of insurance companies and pension fund liquidity, especially for defined-benefit pension plans. However, significant capital and liquidity reserves held at insurance companies and pension funds (OECD, 2023[16]; OECD, 2023[17]) make them less likely to be forced sellers of commercial real estate financial assets at distressed valuations.

High-level policy considerations

This section draws together the major challenges facing risk assessments of commercial real estate and macroprudential policies targeted at banks and non-bank financial intermediaries to limit the buildup of system-wide financial risk. Some policy considerations are then suggested that could strengthen the resilience of the global financial sector.

Significant data gaps undermine accurate assessment of financial risks in the commercial property sector

The complexity, opacity and heterogeneity of the commercial property sector and its financing channels lead to significant data gaps about the property market (e.g. commercial buildings), the exposures of a variety of funding providers and their lending standards. Commercial real estate statistics suffer from data

[15] Commercial equity REITs typically derive their returns from rents paid by tenants and the changes in the valuation of commercial properties. This is not dissimilar to commercial mREITs which derive their returns from the income generated by underlying commercial mortgages in their portfolios.

[16] See Box 1 for more details about the use of leverage by REITs.

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quality issues, with scarce and incomplete data, and a lack of comparability across jurisdictions. Even where comparable data are available, consistent time series information is often limited (ESRB, 2019[18]; ESRB, 2023[19]). The analysis in this paper highlights the challenges in obtaining detailed and harmonised cross-country commercial property price indicators, and breakdowns of commercial real estate exposures by asset classes and type of collateral, both for tangible commercial property and for the commercial real estate financial product holdings by various financial intermediaries.[17] The lack of global harmonised, granular and high-frequency data frameworks and early-warning indicators to capture broader commercial real estate market developments hinders accurate analysis and monitoring of financial stability risks across financial intermediaries and jurisdictions. This makes it difficult for regulators to calibrate effective and efficient policy tools to take action without delays.

A precise definition of commercial property assets would help financial stability monitoring and macroprudential policymaking (Dierick et al., 2017[20]). However, there is no consensus on this. One approach is to follow the national accounts definition, with commercial property assets defined as all non-residential private real estate assets.[18] A potential drawback is that this definition could include buildings that are of limited relevance for financial stability or macroprudential policymaking (such as religious buildings). An alternative approach is to define commercial property assets as any income-producing real estate, either existing or under development, with the exclusion of property owned by end-users, social housing, and buy-to-let housing.[19] This alternative definition encompasses all types of buildings that might raise regulators' concerns about unsustainable credit or asset price developments, or insufficient cash-flow generating capacity to repay the contracted loans. Under this definition, multi-family properties[20], which are considered commercial property assets in the United States, would be included within the scope of financial stability monitoring, as already occurred during the GFC in several advanced economies and more recently in China. The alternative definition of commercial property assets would mean that commercial real estate sector risks are affected by both business and household vulnerabilities, highlighting the need for regulators to use multiple tools in monitoring risks.

A sound risk monitoring framework is required to identify the main threats to financial resilience (OECD, 2021[21]). This would help identify excessive risk-taking behaviour by commercial property developers and investors with systemic implications in order to prevent a price bubble build-up in some credit market segments.[21] Such monitoring requires an array of qualitative and quantitative indicators on the commercial property market, commercial real estate exposures of financial intermediaries and credit standards. Information about physical property exposures may include property type and location, which matter for

[17] There are also challenges in aligning green real estate finance market practices with sustainability goals, due to a lack of common definitions, standards and quality data (OECD, 2022[8]).

[18] This definition is also consistent with the approach used for bank capital requirements in certain jurisdictions (such as the Article 4(1) (75) of the Regulation (EU) No 575/2013 of the European Parliament and of the Council of 26 June 2013 on prudential requirements for credit institutions and investment firms) and the AnaCredit database (i.e. including detailed information on individual bank loans in the euro area).

[19] As suggested by the ESRB recommendation 2019/3 amending the Recommendation ESRB/2016/14 on closing real estate data gaps.

[20] In the United States, smaller multi-family homes are considered as residential property if they have four or fewer units. A conventional loan can be used when investing in a property with four or fewer units. A multi-family property with more than four units must be financed with a commercial loan. Loan financing of residential property with one-to-four units is subject to federal conforming loan limit values (FHFA, 2023[68]). Financing for five units and more multi-family properties is evaluated on a case-by-case basis with amounts exceeding conforming loan limits.

[21] The OECD Policy Framework for Effective and Efficient Financial Regulation states that a sound risk assessment of exposures is essential to identify major vulnerabilities effectively and establish the case for intervention to issue relevant policy recommendations.

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the valuations and yields of property investments. Monitoring the debt and equity financing of commercial real estate investments requires detailed information on both lenders and investors by type (banks, insurance companies, investment funds) and nationality (domestic or foreign). Information about collateral and incomes would also help the monitoring of credit risk pricing and debt sustainability dynamics. Enhanced sustainability disclosure and strengthened green bond and mortgage loan frameworks could also improve the existing tools, methodologies and products to support an orderly climate transition in commercial real estate finance (OECD, 2022[8]; OECD, 2024[9]).

Heterogeneous implementation of macroprudential policy tools in commercial real estate finance across jurisdictions

The use of macroprudential policy tools to mitigate commercial property risks is more challenging than in residential real estate markets due to the complexity and heterogeneity of the commercial property sector, the range of market participants and instruments, and data quality issues related to lending standards for commercial real estate loans (ECB, 2022[22]; ESRB, 2023[19]). Each commercial real estate sub-sector and market segment has distinct characteristics and interconnections with the broader economy. Assessments of collateral value may also be challenging as historical valuation data may not always be readily accessible or still relevant at the time of loan issuance. This is particularly the case in market segments with limited transactions. Verifying a borrower's own resources can also prove challenging due to complex ownership structures and potential reliance on revolving market financing.

The macroprudential toolkit in real estate finance remains primarily bank-based (Box 2). The main targeted policy tools include ex-post capital-based and ex-ante borrower-based measures. Capital-based measures are intended to boost lender resilience – by adding macroprudential capital requirements (e.g. higher risk weights or stricter criteria on risk weights, systemic sectoral capital buffers, stress-testing exercises) to cover systemic risks not reflected in microprudential capital requirements[22] – and support credit intermediation following a shock. Borrower based measures are intended to limit borrowers' excessive indebtedness and ensure solid credit quality and adequate repayment capacities, thereby limiting future non-performing loans and losses (e.g. commercial real estate specific loan-to-value ratio and debt service-to-income ratio). Financial stability would be enhanced if a level playing field was ensured by applying such activity-based tools to all providers of commercial real estate credit that could pose systemic risk concerns, whether banks or not. As all players of systemic relevance are likely to receive government support during times of distress, they should be subject to regulatory oversight to mitigate moral hazard.

[22] Microprudential capital measures for banks typically include restrictions related to loan-to-valuation (LTV) ratios. The LTV ratio is the main risk driver for risk weighting purposes with a distinction between exposures secured by real estate where repayment is or is not materially dependent on rent/sale of the property. Banks are required to consider the quality of the collateral (e.g. adequate valuation, finished property), the collateral's effectiveness (e.g. as legal enforceability, seniority of lien), and other procedural aspects (e.g. required documentation) before assigning the minimum risk weights (BIS, 2017[72]). The risk weights applied for commercial real estate exposures where repayment is materially dependent on cash flows generated by property are the minimum between 60% and the risk-weight of the counterparty for an LTV lower than 60%, and otherwise the risk weight of the counterparty. The risk weight of the counterparty is set at 75% for exposures to individuals, and 85% for exposures to SMEs and other counterparties. The risk weights applied for commercial real estate exposures where repayment is not materially dependent on cash flows generated by property are 70% for an LTV below 60%, 90% for an LTV between 60% and 80%, and 110% for an LTV greater than 80%.

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Box 2. A global review of bank-based commercial real estate macroprudential measures in place

Information on macroprudential policies is available from the International Monetray Fund's integrated macroprudential policy (iMaPP) database (Alam et al., 2019[23]), the Bank for International Settlements and European Systemic Risk Board policy databases, and research by (Deghi, Mok and Tsuruga, 2021[24]) and (ESRB, 2023[19]). The limited number of macroprudential measures targeted on commercial real estate means that there are few assessments of their effectiveness. However, the available evidence suggests that measures can be effective: macro-prudential regulation of leverage may help limit commercial real estate asset price booms by preventing sharp falls in risk premia (Duca and Ling, 2020[25]), and the 2006 US commercial real estate lending guidance for banks with a high concentration in the sector reduced their related lending exposure (Bassett and Marsh, 2017[26]).

At least one policy measure to address risks from commercial real estate exposures was activated in 36 jurisdictions globally over the period 2005 to 2023, with some adjustments performed (either strengthening or relaxation) following buoyant conditions in commercial property markets or adverse shocks (such as the GFC or the pandemic).

Risk-weighted capital measures[23] for banks have been adopted in 23 jurisdictions,[24] with financial stability risks from commercial real estate exposures being considered among the factors for adjusting the countercyclical capital buffer in seven European economies (Belgium, Bulgaria, Czechia, Iceland, Ireland, Norway, and Sweden). While recent countercyclical capital buffer decisions in other jurisdictions compliant with the Basel III accords have not specifically cited commercial real estate dynamics as primary concerns, macroprudential authorities often explicitly mention the need for careful monitoring of commercial real estate exposures due to their significance for financial stability.[25] Such exposures have also influenced changes in the systemic risk buffer rate in 7 European jurisdictions (Austria, Czechia, Finland, Hungary, Liechtenstein, Netherlands, and Norway).

The stress-testing scenarios of many major central banks include a scenario of greater stress in commercial property markets, marked by significant price declines that could potentially erode banks' capital buffers and hinder financing for the real economy (Federal Reserve, 2024[27]; ESBR, 2023[28]). In Sweden, the Finansinspektionen implemented in 2020 an extra capital requirement for banks' commercial real estate lending, prompted by the 2019 stress test results revealing potential credit losses exceeding banks' capital for covering risks in commercial real estate lending amid economic stress (Finansinspektionen, 2019[29]).

[23] For banks using the standardised approach for calculating capital requirements, higher risk weights are set for commercial real estate exposures. For banks using internal models, risk weights are increased by raising the regulatory floor of the loss-given-default parameter of the internal models. Nevertheless, limitations prevail in some jurisdictions where the low-risk weights of institutions in the internal ratings-based approach may be driven by a fall in probability-of-default estimates based on past outturns rather than low loss-given-default estimates. Also, risk weight adjustments may apply to banks' retail customer portfolios secured by commercial property, despite the fact that the majority of banks' commercial real estate exposures are in their corporate portfolios.

[24] The following jurisdictions are included: Benin, Burkina Faso, Cote D'Ivoire, Croatia, Estonia, Finland, Guinea Bissau, India, Ireland, Kuwait, Latvia, Mali, Niger, Norway, Poland, Romania, Senegal, Spain, Sweden, Togo, United Kingdom, and Viet Nam. The African countries are all members of West African Economic and Monetary Union and have adopted the Basel Accords to regulate and supervise banks.

[25] For example in France (Haut Conseil de Stabilité Financière, 2021[64]), Germany (Deutsche Bundesbank, 2024[65]), Japan (Bank of Japan, 2024[66]), and Australia (APRA, 2021[67]).

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Borrower-based measures for commercial real estate exposures have been adopted in 25 jurisdictions.[26] Other loan-based measures have been implemented, such as restrictions on credit institutions' aggregate exposures to commercial real estate in Denmark; loan loss provisions, reserve requirements and a limit on credit growth in India; a dedicated asset management company to purchase distressed commercial real estate portfolio from financial institutions in Hungary; and mandatory disclosure of bank commercial real estate exposures in The Philippines. Tighter managerial controls have been required in the United States for banks with risks concentrated on commercial real estate.

Macroprudential measures to address risks from commercial real estate exposures for non-bank financial intermediaries are scarce and heterogenous across jurisdictions.

- In the pension fund and insurance sectors, risks related to their commercial real estate exposures differ significantly across jurisdictions, reflecting both different business models and practices as well as different market structures. Insurance companies are mainly subject to capital charges for property and concentration risks from individual counterparty exposures but not necessarily for risks from specific sectors and geographical regions (European Parliament, 2021[30]; NAIC, 2024[31]). Pension funds are also subject to targeted regulations for their real estate investments (either direct or indirect) through portfolio, investment and loan limits, and possibly other quantitative investment regulations (OECD, 2023[32]). Nevertheless, the scope and type of regulatory measures varies substantially across jurisdictions.
- Real estate investment funds are subject to liquidity and leverage limits, particularly in Europe (ESRB, 2023[19]; Munoz and Smets, 2023[33]). According to an ESRB survey, measures used include limits on balance sheet leverage and the use of ex-ante (e.g. liquidity buffers) and ex-post liquidity management tools (such as suspension of redemptions, redemption fees or short-term borrowings from real estate funds to settle redemptions). Also, in some European jurisdictions, certain types of real estate funds have to comply with on-balance sheet leverage limits, which correspond to loan-to-value limits for commercial real estate portfolios. In 2022, the Central Bank of Ireland became the first authority to introduce a 60% macroprudential leverage limit (using the debt to assets ratio) and guidance to limit liquidity mismatch for real estate funds (Central Bank of Ireland, 2022[34]).[27] The Central Bank provided a five-year implementation period to allow for the gradual and orderly adjustment of leverage in existing property funds. An eighteen-month implementation period was provided to existing funds as of November 2022 to take appropriate actions in response to the Guidance, which was revised and enhanced in 2023-2024 (Central Bank of Ireland, 2024[35]).
- As with real estate investment funds, REITs are subject to leverage and asset or level risk measures, which are typically governed by specific national regulations and vary across jurisdictions (EPRA, 2023[36]). Annex A provides a comparative summary of the REITs regime in G7 countries. This highlights the heterogeneous key regulatory requirements across G7 economies (such as shareholder and listing requirements, restrictions on investments and activities, leverage and profit distribution obligations).

[26] The following jurisdictions are included: Austria, Belgium, Bulgaria, Croatia, Curacao, Cyprus, Denmark, Estonia, France, Germany, Ireland, Italy, Latvia, Lithuania, Malta, Mauritius, Poland, Portugal, Romania, Singapore, Slovenia, Spain, Sweden, Tanzania, and Türkiye.

[27] As of mid-2022, 160 identified property funds held EUR 22.1 billion in Irish property assets across various market segments, accounting for 35% of Ireland's investable commercial property assets. Their holdings, mainly in Dublin, included office, retail, and residential real estate. In 2020, property assets were allocated as follows: office 37%, retail 26%, residential 15%, and other 23%. Property funds are used as a generic term for real estate investment funds that invest in commercial and/or residential real estate assets.

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Continued monitoring of vulnerabilities from commercial real estate exposures is warranted and additional policy measures could help to minimise financial stability risks (ESRB, 2023[19]; European Central Bank, 2023[12]; IMF, 2024[37]; OECD, 2021[21]). Credit risk exposure from commercial real estate loans for both banks and non-bank financial intermediaries could be mitigated by in-depth assessment and monitoring of the credit quality of commercial real estate portfolios and the adequacy of provisioning, with a particular focus on loans that could be negatively affected by changes in the commercial real estate sectors. Credit providers should already regularly reassess commercial real estate collateral values and the debt-servicing capacity of their borrowers, using prudent assumptions about sectoral developments, and the evolution of rents, vacancy rates and construction costs (ECB, 2024[38]). However more could be done to enforce this. Supervisors should also conduct stress-testing scenarios of large commercial property price declines for all banks with material commercial real estate exposures rather than a subset of the largest banks only, with account taken of indirect exposures to commercial real estate through bank funding of exposed non-bank financial intermediaries.

In the insurance and pension fund sectors, accurate risk assessment of commercial real estate exposures and the potential liquidity mismatches between their funding and their commercial real estate investments is a prerequisite for implementing effective and efficient policy measures. In particular, understanding how collateral for commercial real estate loans and investments is treated, and how risks from commercial real estate exposures are accounted for, including indirect risks from investments in specialist investment funds, would help to improve capital requirements, investment rules and risk management practices.

Measures to address liquidity and leverage risks at open-ended real estate investment funds would help to strengthen their resilience to margin calls and redemptions (FSB, 2023[39]; ECB, 2024[40]). Consistent with policy recommendations from the Financial Stability Board and the International Organization of Securities Commissions (FSB, 2023[41]; IOSCO, 2024[42]), real estate funds should redeem shares less frequently, requiring a better alignment between fund redemption terms and the liquidity of underlying commercial real estate assets. Close monitoring of the risks arising from liquidity mismatch and leverage is also a prerequisite for the effective use of liquidity management tools and leverage limits where necessary.

Further exploratory and conceptual work on borrower-based measures for commercial real estate loans needs to reflect the heterogeneity of the commercial property sector and its financing. This includes how to use multiple tools optimally and quantify the costs and benefits precisely (BIS, 2023[43]). Transparency about the challenges of using macroprudential policies to mitigate risks from commercial real estate exposures will in the long run enhance policymaking even if in the short term it might heighten scrutiny.

To effectively complement entity-based regulatory measures (such as regulations on banks), activity-based measures for commercial real estate lending should be applicable to all credit providers that are interconnected and may pose systemic risk concerns (including for cross-border lending). This idea corresponds with the “congruent principle for financial regulation” put forward by (Metrick and Tarullo, 2021[44]), which states that forms of financial intermediation posing similar risks to financial stability should be regulated with similar stringency, regardless of legal form, chartering identity, or business model. Thus, the amount of systemic risk tolerated in non-bank intermediation should be similar to that implicitly incorporated in prudential regulation of banks, and reflect the same balance between short-term growth and medium-term financial stability. A holistic approach to financial sector risk monitoring and supervision (Acharya, Cetorelli and Tuckman, 2024[45]), recognising the implications of complex relationships between banks, real estate investment trusts and funds, and institutional investors, would help mitigate regulatory arbitrage and negative externalities. This would prevent disruptions in credit provision or defaults that could destabilise banks, other intermediaries, and parts of the real economy.

Enhanced international cooperation is also needed to identify any additional relevant policy instruments and reduce regulatory fragmentation across jurisdictions, thereby enhancing global financial stability.

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Annex A. Comparative summary table of REIT regimes in G7 countries

Country	Enactment year	Minimum share capital	Listing mandatory	Shareholder requirements	Restrictions on investments and activities	Leverage	Profit distribution obligations
Canada	1994	No	Yes (to avoid redemption right of unitholders)	Minimum of 150 unitholders, each of whom holds not less than one block of units and has an aggregate fair market value of no less than CAD 500 Generally, REITs cannot be established or maintained primarily for the benefit of non-residents of Canada	Investing in property (other than real property or an interest in real property) is allowed Acquiring, holding, maintaining, improving, leasing or managing any real property (or interest in real - property) that is the capital property of the trust is allowed Any combination of the foregoing activities	No limitation on leverage specifically, but thin capitalisation rules may apply to limit tax-deductible interest	All income of the REIT for a taxation year is paid or payable to unitholders in the year so that REIT does not incur tax All capital gains are paid out and retain their character as such in the hands of unitholders provided a designation is made by the REIT
France	2003	Required minimum share capital of the listed parent of EUR 15 million	Yes	Shareholders must not hold more than 60% of share capital or voting rights except for subsidiaries of a REIT parent company At the time of the election, 15% of the share capital and voting rights must be held by shareholders, who individually own fewer than 2% Same requirements for foreign shareholders	Principal activity restricted to property acquisition and/or construction with the aim of renting out the property as well as direct or indirect portfolio investments in partnerships Ancillary activities such as property development or brokerage must not exceed 20% of the REIT's assets gross book value	No leverage restrictions but several interest deduction limitation rules may apply	At least 95% of the tax-exempt profits derived from qualifying leasing activities to be distributed before the end of the tax year At least 70% of the capital gains must be distributed before the end of the second tax year 100% of the dividends received from the REIT's subsidiaries that have elected for the REIT regime must be distributed before the end of the tax year

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Country	Enactment year	Minimum share capital	Listing mandatory	Shareholder requirements	Restrictions on investments and activities	Leverage	Profit distribution obligations
Germany	2007	Required minimum share capital of a G-REIT of least EUR 15 million	Yes	15% of the shares must be widely held by at least 6 shareholders (25% at the time of initial public offering) A shareholder is not allowed to own directly 10% or more of the shares or the voting rights of the company Same requirements for foreign shareholders	A minimum of 75% of total assets must be immovable property (excluding pre-Jan 1, 2017), with the primary focus on rental activities or immovable property disposal, and secondary activities, like development, permitted through wholly owned subsidiary, not exceeding 20% of gross earnings At least 75% of its gross earnings must derive from rental, leasing, letting and disposal of immovable property	The equity must equal at least 45% of the total asset value of immovable property	90% of the net income must be distributed until the end of the following business year Up to half of the proceeds from capital gains can be transferred to a reserve (distributable profits to be reduced accordingly)
Italy	2007	Ordinary listing requirements in respect of share capital are applicable to REITs	Yes	At least 25% of the shares must be widely held A single shareholder is not allowed to own more than 60% of voting rights and profit participation rights Same requirements for foreign shareholders	Main activity (at least 80%) must consist of (1) real estate properties to be leased (2) participations accounted as fixed assets in REITs and Italian real estate investment funds whose real estate assets held for lease or participation in those are at least 80% of the total assets. Assets under construction or subject to renovation works included if intended to be leased At least 80% of RIETs income must be (1) proceeds from lease activity (2) dividends from leasing activity participations in REITs and qualifying real estate investment funds (3) capital gains realized on the disposal of real estate properties held for lease or of participations in REITs and qualifying real estate funds	The leverage cannot exceed the ratio resulting from the company's by-law	70% of net profits deriving from the leasing activity must be distributed annually At least 50% of net capital gains realized on the disposal of real estate properties must be distributed within the two years following the disposal
Japan	2000	Required minimum share capital amounts of JPY 100 million	No	No requirements under the Investment Trust Law Special shareholder conditions in order to deduct dividend distribution under the Special Taxation Measures Law	A J-REIT is merely an asset holding vehicle Investment primarily in Qualified Assets	No gearing loan-to-value limit under applicable law May only obtain loans from Qualified Institutional Investors	Greater than 90% of distributable profits under the Special Taxation Measures Law Same as ordinary income for capital gains distribution

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United Kingdom	2006	Normal listing requirements apply for the respective stock exchange where the REIT shares are listed	Not mandatory if at least 70% of the shares are owned by one or more institutional investor	A UK REIT cannot be a 'close company', e.g. controlled by five or fewer shareholders if at least 35% of the shares are owned by the public, not considered as a 'close company' Potential penalties if a single corporate shareholder owns 10% or more of the shares/voting rights No restrictions for foreign shareholders	At least 75% of a REIT's assets must be used in the property rental business Development by the UK REIT for investment on its own account is permitted Other activities are permitted subject to these restrictions At least 75% of a REIT's net profits must be derived from the property rental business	Interest expense is limited by the financing cost ratio	90% of tax-property rental profits must be distributed within 12 months of the end of the year Where a REIT invests in another REIT, 100% of property income distribution from other REITs must be distributed within 12 months of the end of the accounting period Capital gains are not included in the distribution obligation
United States	1960	No minimum share capital requirement	No	At least 100 shareholders Five or fewer individuals or foundations may not hold more than 50% of the share No restriction on foreign shareholders	At least 75% of its assets must be real estate, government securities or cash At least 75% of the gross income must be derived from real estate property rental or interest on real estate property mortgages At least 95% of the gross income must come from a combination of real estate-related sources and passive sources, such as dividends and interest Cannot own more than 10% of another corporations stock other than in another REIT or a taxable REIT subsidiary (ownership of a 100% owned qualified REIT subsidiary is ignored) No more than 5% of the value of its assets can be represented by securities of any one issuer other than another REIT or a taxable REIT subsidiary (ownership of a 100% owned qualified REIT subsidiary is ignored) Cannot own more than 20% of its assets in securities of one or more taxable REIT subsidiaries	No legal restrictions	At least 90% of its taxable ordinary income Capital gains not required to be distributed

Source: EPRA Global REIT Survey 2023, OECD.

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Annex B. Methodology for commercial real estate market-based credit data collection and calculations

As commercial real estate investment statistics suffer from both data quality (scarce and incomplete data) and operative issues (lack of comparability across countries and – even where comparable data are available – insufficiently long time series), it is difficult to obtain a comprehensive assessment of vulnerabilities in the sector across jurisdictions globally.

The new datasets assembled for this paper are intended to provide global coverage for various credit market products, real estate investment trusts (REITs), and commercial property developers. Granular historical market and financial statement information at the firm or deal level provides a clear view of business-oriented property development and services, excludes income-generating residential real estate (such as US multi-family units), and aligns with the common definition of commercial real estate in national accounts.

Balance sheet information for listed REITs and property developers

Listed commercial equity and mortgage REITs

The unbalanced panel dataset used in this paper contains financial statement information for 801 listed commercial and residential equity REITs in 21 countries between 2000 and 2023. The sectoral distribution of REITs is based on the Industry Classification Standard (GICS) industry name. 719 commercial equity REITs operate in retail, office, lodging, healthcare and industrial sectors. REITs operating in specialised or diversified real estate sectors are also included in the group of commercial REITs.[28] Remaining REITs are operating in the residential real estate sector and are excluded from the dataset. Financial statement information has been collected for an unbalanced panel of 53 listed US mREITs, of which 22 are commercial mREITs, between 2005 and 2023.

Property developers

A property development company manages new construction projects, building and marketing residential or commercial properties. It often oversees all project phases and may own its own construction company.

The unbalanced panel dataset used in this paper contains financial statement information for 2146 property developers in 86 countries between 2006 and 2023. Property developers have been identified in the LSEG database using the Reference Data Business Classification (TRBC) industry name and are performing real estate rental, development and operations or real estate services. More detailed industry classification information is not available to distinguish between commercial and residential property developers.

[28] Diversified equity REITs invest in various commercial property segments and possibly also in residential real estate, while specialised REITs focus on particular commercial and/or residential property segments. However, more detailed industry classification information is not available to distinguish between commercial versus residential specialised REITs.

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In the two datasets (for REITs and property developers), the country classification is based on the International Monetary Fund's country classification by considering the country of incorporation of the REIT or the property developer. Data are adjusted into 2023 USD prices using the US consumer price index.

Bonds, mortgage-backed securities and syndicated loans

Bonds from commercial property developers and equity REITs

The dataset used in this paper includes 28,926 bonds issued by property developers collected from LSEG across 80 countries between 1980 to 2023, of which 23956 bonds issued by commercial and diversified or specialised property developers. For REITs (i.e. residential and commercial equity REITs and mREITs), the database comprises 5119 bonds issued across 50 countries from 1990 to 2023, of which 4513 bonds issued by commercial and diversified or specialised equity REITs (i.e. investing in retail, office, lodging, healthcare, industrial, diversified and specialised property sectors).[29] Convertible bonds, deals that were registered but not consummated, preferred shares, sukuk bonds, bonds with an original maturity less than or equal to 1 year or an issue size smaller than USD 1 million are excluded from the dataset. The industry classification is based on the sector of the issuer.

For property developers, bonds issued by commercial property developers have been identified based on the following issuer TRBC sectors: office real estate rental and development, retail real estate rental and development, office real estate services, industrial real estate rental and development, industrial real estate services, and retail real estate services. Bonds issued by residential property developers have been identified based on the following issuer TRBC sectors (but are excluded from this dataset): residential real estate rental and development, and residential real estate services. Bonds issued by diversified or specialised developers[30] have been identified based on the following issuer TRBC sectors: real estate rental, development and operations, and real estate services.

For equity REITs, bonds issued by commercial equity REITs have been identified based on the following issuer TRBC sectors: commercial REITs, healthcare REITs, hospitality REITs, industrial REITs, office REITs, retail REITs, self-storage REITs, and timber REITs. Bonds issued by residential equity REITs are classified under the TRBC sector for residential REITs and are excluded from this dataset. Bonds issued by diversified or specialised equity REITs have been identified based on the following issuer TRBC sectors: diversified REITs, and specialized REITs.

Commercial real estate sustainable bonds

The dataset of sustainable bonds contains deal-level information of 1047 bonds from issuers operating in the real estate sector (under the TBRC sector for real estate rental, development and operations) in 68 countries from 2012 to 2023, of which 944 bonds issued by commercial and diversified or specialised property developers. Bonds have been classified by type using the sustainable bond type identifier from LSEG. Green bonds include Climate Bond Initiative aligned, certified and self-labelled green bonds. Sustainability linked bonds link the bond's coupon to the achievement of environmental, social or governance targets. A remaining category includes social bonds.

Sustainable bonds issued by commercial property developers have been identified based on the following issuer TRBC sectors: commercial real estate, office real estate rental and development, retail real estate

[29] A very small number of bonds have been collected for mREITs that have been excluded from the dataset.

[30] Same definition and data challenges as encountered with equity REITs.

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rental and development, office real estate services, industrial real estate rental and development, industrial real estate services, and retail real estate services.

Sustainable bonds issued by residential property developers have been identified based on the following issuer TRBC sectors and are excluded from this dataset: residential real estate, residential real estate rental, and development and residential real estate services.

Sustainable bonds issued by diversified or specialised property developers have been identified based on the following issuer TRBC sectors: real estate rental and development, and real estate services.

LSEG data contains both Regulation S and Rule 144A sustainable bonds. The calculations presented take account of this factor, and an exercise to eliminate the duplication when a single bond was issued both under Regulation S and Rule 144A was performed. Additionally, bonds cancelled on or before Initial settlement are excluded. The final dataset includes 944 bonds issued by commercial and diversified or specialised property developers.

Commercial mortgage-backed securities

The dataset includes 33,357 mortgage-backed securities (MBS) collected from LSEG issued in 64 countries from 1980 to 2023, of which 19790 are commercial and diversified or specialised real estate MBS. MBS that were registered but not consummated are excluded from the dataset. Commercial and residential MBS are identified based on the TRBC industry of the issuer and possible relevant commercial and residential real estate acronym in the issuer name.

Commercial MBS identified in the dataset (bonds) include:

- MBS issued by Farmer Mac.
- MBS with an issuer operating in a commercial non-financial TRBC industry sector.
- MBS with an issuer name containing the following commercial real estate acronyms (and their translation in other languages) such as “commercial”, “CMO”, “office”, “retail”, “shopping”, “hotel”, “lodging”, “hospitality”, “industrial”, “logistic”, “self-storage”, “timber”, and “CMBS”.

This report classifies multi-family MBS as residential MBS, aligning with the common global definition of commercial real estate in national accounts.

Residential MBS identified in the dataset are excluded. These include:

- MBS issued by agencies (including Fannie Mae, Freddie Mac, Ginnie Mae, US Department of Veterans Affairs, US Government National Mortgage Association and US Federal National Mortgage Association).
- MBS with an issuer operating in TRBC consumer and governmental services sectors.
- MBS with an issuer name containing the following residential real estate acronym (and their translation in other languages) such as “housing”, “home”, “residential, “RMBS”, and “multi-family”.

Diversified or specialised MBS included in the dataset contain all MBS not identified above and with an issuer operating in the following TRBC industries: financial and investment services specialty or financial services diversified.

In these bonds and MBS datasets, the country classification is based on the International Monetary Fund’s country classification by considering the country of domicile of the issuer. Data are adjusted into 2023 USD prices using the US consumer price index. Bond values correspond to the gross proceeds (i.e., the amount paid by investors to acquire the bonds) in most cases. Where the information on the gross proceeds is not available, the original amount issued is used. When calculating the outstanding amount of corporate bonds in a given year, issues that are no longer outstanding (due to being redeemed earlier than their maturity)

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are deducted. The early redemption data are obtained from LSEG WS and cover bonds that have been redeemed early due to being repaid via final default distribution, called, liquidated, put or repurchased.

Syndicated loans to property developers

The global database includes 22,583 syndicated loan deals issued to property developers across 73 countries between 1983 to 2023.[31] The industry classification is based on TRBC industry from LSEG, but does not include detailed information to distinguish between commercial and residential property developers. Leveraged loans are identified based on the loan status from LSEG. Registered but not consummated loan deals, those with an original maturity less than or equal to 1 year, and those for a loan smaller than USD 1 million are excluded from the dataset. The country classification is based on the IMF's country classification by considering the country of domicile of the borrower.

The outstanding amount is calculated based on loan issuance but excludes the value of drawn and undrawn revolving credit facilities. A linear amortisation schedule is postulated for term loans A[32] and other amortising loans (i.e., mortgages, equipment, construction, commercial loans). All other term loans are not amortised as they are repayable at maturity. To account for loan re-financing, a 40% early repayment ratio is used.

The debt composition by credit quality

The composition of debt (reported on the liability side of the balance sheet) by level of credit quality is determined based on leverage (measured by the ratio of debt to EBITDA[33]) and interest coverage (measured by the ratio of EBITDA to interest expense) ratios. The interest coverage ratio is an indicator of the ability of a company to make interest payments using internal cash flows. As such, numerous studies suggest that the level of the interest coverage ratio can be an important indicator of financial distress under high-interest rates. The criteria for decomposing debt by credit quality are based on previous OECD work (Roulet, 2020[46])

Corporates with "sound credit risk profile" have a positive leverage ratio lower than 5 and an interest coverage ratio higher than 2. Leveraged corporates are companies with a leverage ratio higher than 5 or with a negative leverage ratio (due to negative EBITDA). Along these, "at risk" corporates have an interest coverage ratio between 1 and 2, "distressed" corporates an interest coverage ratio lower than 1, and "moderately leveraged" an interest coverage ratio higher than 2.

[31] Global syndicated loan issuance for commercial equity REITs represents a very small proportion of global issuance (accounting for less than 1% to 2% of global issuance from 2000 to 2023) and is not shown in the report.

[32] A senior term loan that usually matures within five to six years. Term loan A (TLA) tranches typically amortize, with the borrower having to repay an amount of the term loan A each year equal to between 5% and 20% of the initial principal amount of the loan. The amount of the amortization payment generally increases during the term of the loan, with the percentage amount starting low and increasing each year until the maturity date.

[33] EBITDA stands for earnings before interest, tax, depreciation and amortisation.

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